dirac16complex00 in the primordial gravitational field

The semi-classical Pin(4,4) spinor field: Lagrangian, field equations, non-triviality [2], energy-momentum tensor, equations of state and the field equations for a4[x4] (Revision record)

dirac16complex project (with Claude Opus 5.5)

September 2026

Contents

1 Scope, sources and conventions **3**

2 The field dirac16complex00 and its symmetry group **5**

3 Coupling to the primordial field: vielbein and canonical spin connection **6**

4 The Lagrangian **8**

5 The exact field equations **9**

6 Non-triviality [2]: theorem and proof **12**

7 Self-consistency of the field equations **15**

8 The energy-momentum tensor (classical bilinears) **17**

9 Kinetic energy, potential energy, energy density and pressures **19**

10 Equations of state **20**

11 The field equations for a4[x4] **22**

12 dirac16complex00 as the source of the a4 equations **25**

13 Pairing theorems for dirac16complex00 (summary) **28**

14 Check index **29**

15 What is not claimed **35**

16 Reproduction **36**

Abstract

This document records, for the author's primordial gravitational field of `Revision/SPEC.md` section 1, the field theory of dirac16complex00: a field Φ with 16 complex commuting components that transform as a spinor of Pin(4,4) (a set of 16 scalar fields that transform as a Pin(4,4) spinor), treated as a classical ("semi-classical") field. It gives the Lagrangian and its coupling to gravity through the vielbein and the canonical spin connection, the exact Euler-Lagrange equations (covariant, explicit in the metric, and as 16 component equations), the non-triviality theorem [2] with its proof and its exact scope (in the diagonal vielbein the gravitational term of the field equation is $\gamma^\mu\Omega_\mu = 3H\gamma^{(x_8)}$, nonzero for every $H > 0$; this value depends on the frame and on the field variables, while Ω_μ itself vanishes in no frame because the metric is curved), the self-consistency of the equations and its limits (no well-posed Cauchy problem for data that depend on the extra times; boundary terms at $z = \pi/2$), the energy-momentum tensor as a classical bilinear, its kinetic and potential parts, the energy density, the pressures and the equations of state, and the Einstein-Lovelock field equations for $a_4(x_4)$ with dirac16complex00 as the source. Every formula is an output of Revision code (Wolfram and an independent sympy checker for each part); each section ends with the names of the checks that verify it, and the check index collects them. Nothing is taken from the earlier stages of the repository. The classical energy of dirac16complex00 is unbounded below already for $U = 0$ (positive-frequency modes of negative energy) and its charge is indefinite and only locally conserved (the charge of a slice is constant only if no charge flows through the brane $z = \pi/2$, where this record imposes no boundary condition). What the record does not establish is stated in the section "What is not claimed". Added on 2026-10-08: the verdict of the dark-sector investigation of Hypothesis00 (section 10, from `Revision/docs/DARK_SECTOR_HYPOTHESES`) and pointers to the other Revision documents (section 1).

1 Scope, sources and conventions

Sources. Every formula and number below is read from these Revision outputs (all produced by Revision code, all checks passing):

part	**Wolfram record**	**independent sympy record**
gammas, Pin(4,4), Spin(4,4) (SPEC section 2)	`Revision/algebra /reports/wolfram -algebra.json`, fixture `Revision/algebra/gammas. json`	`Revision/algebra /reports/python-algebra. json`
geometry, Lagrangian, field equations, energy-momentum tensor (SPEC sections 1-4)	`Revision/theory/reports/ wolfram-field-th eory.json`, formulas `Revision/theory/ field-theory.json`	`Revision/theory/reports/ python-field-the ory.json`
field equations for $a_4(x_4)$ (SPEC section 5)	`Revision/field_e quations_a4/repo rts/wolfram-a4-r eport.json`, formulas `Revision/field_e quations_a4/a4-e quations.json`	`Revision/field_e quations_a4/repo rts/python-a4-re port.json`

part	Wolfram record	independent sympy record
pairing theorems (SPEC section 9)	`Revision/pairing/reports/wolfram-pairing.json`, theorems `Revision/pairing/pairing-theory.json`	`Revision/pairing/reports/python-pairing.json`
scope of the statements: frame dependence, boundary terms, growth, sign of the energy	`Revision/theory/reports/wolfram-scope.json`	`Revision/theory/reports/python-scope.json`

The Wolfram and the sympy records are written independently (no shared code); each sympy record re-derives the statements and compares them with the Wolfram record. The theory records verify every statement for both statistics: the checks whose names end in `_C` (Wolfram) or start with `commuting_` (sympy) are the ones for the commuting field dirac16complex00 of this document; the Grassmann versions belong to dirac16complex (`Revision/docs/DIRAC16COMPLEX_FIELD_THEORY`).

Related Revision documents. The generalized Kronecker delta and the Lovelock tensors $P_{(1)}$, $P_{(2)}$, $P_{(3)}$ behind the field equations for a_4 of section 11 are described in `Revision/docs/LOVELOCK_GKD`. Hypothesis00 (SPEC section 8) is investigated in `Revision/docs/DARK_SECTOR_HYPOTHESES`, whose verdict for dirac16complex00 is summarised at the end of section 10; its two dirac16complex00 records are `Revision/dark_sector/dirac16complex00/reports/python-derive-eos.json` (the derivations and the models; 49 of 49 checks pass) and `Revision/dark_sector/dirac16complex00/reports/python-independent-numerics.json` (an independent implementation; 28 of 28 checks pass). The pairing theorems for both fields are in `Revision/docs/PAIR_CREATION_PROOFS`. The Kohn-Sham model of `Revision/docs/KOHN_SHAM_DEFLATING_FIELD` belongs to dirac16complex and is not used here.

Coordinates (the author's names). x_1, x_2, x_3 are ordinary 3-space (inflating, scale factor $e^{a_4}\sin^{1/6} z$); x_4 is the time; x_5, x_6, x_7 are the three extra times, time-like, which deflate exponentially (scale factor $e^{-a_4}\sin^{1/6} z$ with a_4 increasing); x_8 is the hidden space direction. Throughout $z = 6Hx_8 \in (0, \pi/2)$, $H > 0$ is the author's constant and $a_4' = da_4/dx_4$. In formulas the index i runs over x_1, x_2, x_3 and t over x_5, x_6, x_7.

The metric (SPEC section 1, exactly):

```
g = {{E^(2 a4[x4]) Sin[6 H x8]^(1/3),0,0,0,0,0,0,0},
     {0,E^(2 a4[x4]) Sin[6 H x8]^(1/3),0,0,0,0,0,0},
     {0,0,E^(2 a4[x4]) Sin[6 H x8]^(1/3),0,0,0,0,0},
     {0,0,0,-1,0,0,0,0},
     {0,0,0,0,-E^(-2 a4[x4]) Sin[6 H x8]^(1/3),0,0,0},
     {0,0,0,0,0,-E^(-2 a4[x4]) Sin[6 H x8]^(1/3),0,0},
     {0,0,0,0,0,0,-E^(-2 a4[x4]) Sin[6 H x8]^(1/3),0},
     {0,0,0,0,0,0,0,Cot[6 H x8]^2}}
```

Signature (4,4): space-like x_1, x_2, x_3, x_8, time-like x_4, x_5, x_6, x_7. The volume factor is

$$\sqrt{|g|} = \cos z,$$

independent of x_4: the inflation e^{3a_4} of 3-space is compensated by the deflation e^{-3a_4} of the extra times. $H = 0$ is not a metric of the family (a degenerate limit: $g_{11}, g_{55} \to 0$ and $g_{88} \to \infty$); every statement assumes $H > 0$ and $0 < z < \pi/2$.

Spinor notation. $\bar{\Phi} = \Phi^\dagger C$, $S = \bar{\Phi}\Phi$, $U(S)$ a real potential ($U = \frac{\lambda}{2}S^2$ where stated), $V = m + U'(S)$. Frame gammas are written $\gamma^{(x_a)}$, coordinate gammas $\gamma^\mu = e^\mu{}_a \gamma^{(a)}$. The theory records write Ψ for both statistics; here Φ denotes the commuting field.

Labels. "Exact" means verified by exact symbolic computation (no floating point) in the named checks; "interpretation" marks a reading that is not itself a computed statement.

statement	Wolfram checks	sympy checks
the metric, its vielbein and $\sqrt{\lvert g\rvert} = \cos z$	`metric_is_the_authors`, `sqrt_det_g_is_cos_z`	`metric_from_viel bein_equals_SPEC`, `sqrt_det_g_equals_cos_z`
$H = 0$ is a degenerate limit	`degenerate_at_H_0`	

2 The field dirac16complex00 and its symmetry group

dirac16complex00 is a field Φ with 16 complex commuting components (16 scalar fields) that transform as a Pin(4,4) spinor. It is a classical field; it is not quantised (SPEC section 6: only dirac16complex, the anticommuting field, is quantised canonically). SPEC section 3 calls it the analogue of Dirac's 1928 wave function and the author calls it semi-classical; both are interpretations, and the record shows two exact differences from Dirac's wave function: its charge $Q = \int \cos z \, \Phi^\dagger B \Phi \, d^7x$ (locally conserved, section 7) is an indefinite form, and its classical energy is unbounded below already for $U = 0$ in the good sector (section 9).

The gammas. The real 16×16 matrices are the author's T16, rebuilt in Revision code from the author's formulas (the tau matrices and their blocks), with the frame map $\gamma^{(x_8)} = T16[0]$, $\gamma^{(x_1..x_3)} = T16[1..3]$, $\gamma^{(x_4)} = T16[4]$ and $\gamma^{(x_5..x_7)} = T16[5..7]$. Verified exactly:

- $\{\gamma^{(a)}, \gamma^{(b)}\} = 2\eta^{ab}$ with $\eta = \mathrm{diag}(+1,+1,+1,-1,-1,-1,-1,+1)$ in the order $x_1 \dots x_8$; every $\gamma^{(a)}$ is a real signed permutation matrix; the space-like ones are symmetric, the time-like ones antisymmetric.
- $C = \gamma^{(x_8)}\gamma^{(x_1)}\gamma^{(x_2)}\gamma^{(x_3)}$ is real symmetric, $C^2 = 1$, and $C\gamma^{(a)}$ is real antisymmetric for every a; $C = \mathrm{diag}(-\sigma, \sigma)$ with the notebook's σ.
- $\Gamma = \gamma^{(x_8)}\gamma^{(x_1)} \dots \gamma^{(x_7)} = \mathrm{diag}(-I_8, I_8)$, $\Gamma^2 = 1$, and Γ anticommutes with every $\gamma^{(a)}$.
- $B = -iC\gamma^{(x_4)}$ is Hermitian, $B^2 = 1$, with eigenvalues $+1$ and -1 eight times each: signature (8,8).
- $S^{ab} = \frac{1}{4}[\gamma^{(a)}, \gamma^{(b)}]$ satisfy the so(4,4) relations; $(S^{ab})^T C + C S^{ab} = 0$, so $S = \bar{\Phi}\Phi$ is Spin(4,4)-invariant; $[\Gamma, S^{ab}] = 0$.

Pin(4,4) and Spin(4,4). The 16-dimensional representation is irreducible under Pin(4,4) (commutant dimension 1: the 256 Clifford products span all of $M_{16}(\mathbb{R})$). Under Spin(4,4) it splits into the

two chiral halves $\Gamma = -1$ and $\Gamma = +1$, each irreducible (commutant dimension 1 on each half), and the two 8-dimensional representations are inequivalent (intertwiner dimension 0; the commutant of all S^{ab} on the 16 has dimension 2, spanned by the chiral projectors). Every reflection $\gamma^{(a)}$ exchanges the halves. Pin(4,4) is the double cover of O(4,4), Spin(4,4) of SO(4,4).

Consequence used below. Since $C^2 = 1$ and $C = \mathrm{diag}(-\sigma, \sigma)$ is not ± 1, C has both eigenvalues $+1$ and -1: the bilinear $S = \Phi^\dagger C \Phi$ is an indefinite Hermitian form and takes both signs.

statement	**Wolfram checks (algebra)**	**sympy checks (algebra)**
Clifford relation, reality, symmetry pattern	`Clifford_relation`, `reality`, `symmetry_pattern`, `eta_in_author_order`, `coordinate_map`	`clifford_relation`, `clifford_relation_sympy`, `reality_signed_permutations`, `symmetry_pattern`
C, Γ, B	`C_real_symmetric`, `C_squared_identity`, `C_gamma_real_antisymmetric`, `Gamma_diag`, `B_Hermitian`, `B_squared_identity`, `B_signature_8_8`	`C_equals_notebook_sigma16`, `C_real_symmetric_involution`, `C_gamma_antisymmetric`, `chirality_diag`, `chirality_anticommutes`, `B_hermitian_involution_signature`
so(4,4), invariance of S	`S_Lorentz_algebra`, `S_preserves_C`, `S_commutes_with_Gamma`	`S_lorentz_algebra`, `S_preserves_C_and_commutes_with_Gamma`
Pin(4,4) irreducible; Spin(4,4): two inequivalent irreducible halves	`Pin44_irreducible_commutant_dim_1`, `Spin44_commutant_dim_2_chiral_projectors`, `chiral_halves_irreducible`, `chiral_halves_inequivalent_intertwiners_0`, `Spin_Pin_reflection_swaps_halves`	`clifford_products_span_M16`, `pin_commutant_dimension_1`, `even_products_span_M8_plus_M8`, `spin_commutant_dimension_2`, `spin_halves_irreducible`, `spin_halves_inequivalent`, `reflections_exchange_halves`
the fixture equals the independent construction		`fixture_comparison_gammas_json`, `gammas_json_equals_python_construction`

3 Coupling to the primordial field: vielbein and canonical spin connection

Vielbein. The diagonal vielbein $e^a{}_\mu = f_a \delta^a_\mu$ with

$$f_{x_1} = f_{x_2} = f_{x_3} = e^{a_4} \sin^{1/6} z, \qquad f_{x_4} = 1,$$
$$f_{x_5} = f_{x_6} = f_{x_7} = e^{-a_4} \sin^{1/6} z, \qquad f_{x_8} = \cot z$$

reproduces the author's metric entry by entry, $g = e^T \eta e$, with inverse ${e_a}^\mu = \delta^\mu_a / f_a$ and $\sqrt{|g|} = f_1 \cdots f_8 = \cos z$.

Levi-Civita connection. The 25 independent nonzero Christoffel symbols ($m \le n$) are, for $i = x_1, x_2, x_3$ and $t = x_5, x_6, x_7$ (no sum),

$$
\begin{aligned}
&{\Gamma^{x_i}}_{x_i x_4} = a_4', \qquad {\Gamma^{x_i}}_{x_i x_8} = H \cot z, \qquad {\Gamma^{x_4}}_{x_i x_i} = e^{2a_4} \sin^{1/3} z \, a_4', \\
&{\Gamma^{x_8}}_{x_i x_i} = -e^{2a_4} H \sec z \, \sin^{4/3} z, \\
&{\Gamma^{x_t}}_{x_4 x_t} = -a_4', \qquad {\Gamma^{x_t}}_{x_t x_8} = H \cot z, \qquad {\Gamma^{x_4}}_{x_t x_t} = e^{-2a_4} \sin^{1/3} z \, a_4', \\
&{\Gamma^{x_8}}_{x_t x_t} = e^{-2a_4} H \sec z \, \sin^{4/3} z, \qquad {\Gamma^{x_8}}_{x_8 x_8} = -12H \csc(12 H x_8).
\end{aligned}
$$

Canonical spin connection. ${\omega_\mu}^a{}_b = {e^a}_\nu(\partial_\mu {e_b}^\nu + {\Gamma^\nu}_{\mu\lambda} {e_b}^\lambda)$ satisfies the vielbein postulate $\partial_\mu {e^a}_\nu - {\Gamma^\lambda}_{\mu\nu} {e^a}_\lambda + {\omega_\mu}^a{}_b {e^b}_\nu = 0$ for all 512 components, and $\omega_{\mu ab} = \eta_{ac} {\omega_\mu}^c{}_b = -\omega_{\mu ba}$. Exactly 12 components with $a < b$ are nonzero:

$$
\begin{aligned}
&\omega_{x_i,(x_i)(x_4)} = a_4' e^{a_4} \sin^{1/6} z, \qquad \omega_{x_i,(x_i)(x_8)} = H e^{a_4} \sin^{1/6} z, \\
&\omega_{x_t,(x_4)(x_t)} = -a_4' e^{-a_4} \sin^{1/6} z, \qquad \omega_{x_t,(x_t)(x_8)} = -H e^{-a_4} \sin^{1/6} z,
\end{aligned}
$$

and $\omega_{x_4} = \omega_{x_8} = 0$. Each component is a_4' or H times a factor that never vanishes on the patch.

Spinor connection and covariant derivative. $\Omega_\mu = \frac{1}{2} \omega_{\mu ab} S^{ab}$:

$$
\begin{aligned}
&\Omega_{x_i} = \tfrac{1}{2} e^{a_4} \sin^{1/6} z \, \big(a_4' \, \gamma^{(x_i)} \gamma^{(x_4)} + H \, \gamma^{(x_i)} \gamma^{(x_8)}\big), \qquad i = x_1, x_2, x_3, \\
&\Omega_{x_t} = -\tfrac{1}{2} e^{-a_4} \sin^{1/6} z \, \big(a_4' \, \gamma^{(x_4)} \gamma^{(x_t)} + H \, \gamma^{(x_t)} \gamma^{(x_8)}\big), \qquad t = x_5, x_6, x_7, \\
&\Omega_{x_4} = \Omega_{x_8} = 0,
\end{aligned}
$$

$$
D_\mu \Phi = \partial_\mu \Phi + \Omega_\mu \Phi, \qquad D_\mu \bar{\Phi} = \partial_\mu \bar{\Phi} - \bar{\Phi} \, \Omega_\mu, \qquad \gamma^\mu = {e^\mu}_a \gamma^{(a)}.
$$

This is the canonical connection: it makes the gammas covariantly constant,

$$
D_\mu \gamma^\nu = \partial_\mu \gamma^\nu + {\Gamma^\nu}_{\mu\lambda} \gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0
$$

for all 64 pairs (μ, ν). It is built from the mixed ${\omega_\mu}^a{}_b$ lowered with η (SPEC section 3), not from a contraction of the mixed components without the metric factor.

Curvature. ${R^\mu}_\nu$ is diagonal with

$$
{R^{x_i}}_{x_i} = a_4'' - 6H^2, \quad {R^{x_4}}_{x_4} = 6(a_4')^2, \quad {R^{x_t}}_{x_t} = -a_4'' - 6H^2, \quad {R^{x_8}}_{x_8} = -6H^2,
$$

$$
R = 6\big((a_4')^2 - 7H^2\big).
$$

Since ${R^{x_8}}_{x_8} = -6H^2 < 0$ for every $H > 0$ and every function a_4, the metric is curved for every member of the family, a_4 constant included; all Riemann components vanish only in the formal limit $a_4' = a_4'' = 0$, $H = 0$ (flat 4+4), which is outside the family.

statement	Wolfram checks (theory)	sympy checks (theory)
vielbein and its inverse	`metric_is_the_authors`, `vielbein_inverse`	`metric_from_vielbein_equals_SPEC`
25 Christoffel symbols	`christoffel_count`	`christoffel_symmetric_metric_compatible`
vielbein postulate, antisymmetry, the 12 components	`vielbein_postulate`, `omega_antisymmetric`, `omega_components`	`vielbein_postulate`, `spin_connection_antisymmetric`
Ω_μ; $\Omega_{x_4} = \Omega_{x_8} = 0$	`Omega_components`	`Omega_x4_and_Omega_x8_vanish`
covariant constancy of the gammas	`gamma_covariantly_constant`	`covariant_constancy_D_mu_gamma_nu`
Ricci tensor and scalar; curved for every $H > 0$	`ricci_mixed_components`, `ricci_scalar`, `never_flat_for_H_positive`	`curvature_nonzero_flat_only_formally`

4 The Lagrangian

The Lagrangian density (SPEC section 3; the same form as for dirac16complex, only the statistics differ):

$$\mathcal{L} = \sqrt{|g|}\,\Big[\tfrac{1}{2}(\bar\Phi\gamma^\mu D_\mu\Phi - (D_\mu\bar\Phi)\gamma^\mu\Phi) - m\,S - U(S)\Big], \qquad S = \bar\Phi\Phi = \Phi^\dagger C\Phi,$$

with an explicit mass term linear in m and quadratic in the spinor, and $U(S) = \frac{\lambda}{2}S^2$ where stated (a general real U elsewhere). Proved exactly for the commuting field:

1. $\mathcal{L}$ is real, $\mathcal{L}^* = \mathcal{L}$: C real symmetric, $C\gamma^{(a)}$ real antisymmetric and $(S^{ab})^T C + CS^{ab} = 0$ make the symmetrised kinetic term and $\bar\Phi\Phi$ real without a factor i (the sympy record checks it on 408 monomials, with the control that the unsymmetrised form is not real off shell).
2. $\mathcal{L}$ differs from the unsymmetrised form by a total divergence: $\mathcal{L} = \sqrt{|g|}\,[\bar\Phi\gamma^\mu D_\mu\Phi - mS - U] - \frac{1}{2}\partial_\mu(\sqrt{|g|}\,\bar\Phi\gamma^\mu\Phi)$.
3. For this diagonal metric $\{\gamma^\mu, \Omega_\mu\} = 0$ for each μ separately, so $\bar\Phi\{\gamma^\mu, \Omega_\mu\}\Phi = 0$ and the spin connection drops out of $\mathcal{L}$ itself (next display).

$$\mathcal{L} = \cos z\,\Big[\tfrac{1}{2}\sum_a \frac{1}{f_a}(\bar\Phi\gamma^{(a)}\partial_a\Phi - \partial_a\bar\Phi\,\gamma^{(a)}\Phi) - mS - U(S)\Big].$$

The canonical spin connection reaches the field equations only through the identity

$$\partial_\mu(\sqrt{|g|}\,\gamma^\mu) = 2\sqrt{|g|}\,\gamma^\mu\Omega_\mu$$

in the variation of the derivative terms (section 6). Consequently the Euler-Lagrange equations of $\mathcal{L}$ are exactly those of the connection-free symmetric Lagrangian (the display above), and the surviving term is its half-density (volume and vielbein divergence) term.

Negative control (why the Dirac-type form is used). The notebook's real Majorana-type Lagrangian $L_g = \sqrt{|g|}\,\Theta^T C\gamma^\mu D_\mu \Theta$ with 16 real anticommuting components is a total derivative,

$$L_g = \partial_\mu\big(\tfrac{1}{2}\sqrt{|g|}\,\Theta^T C\gamma^\mu\Theta\big),$$

and has no field equations (all Euler-Lagrange expressions vanish identically). For real commuting components the same L_g is not a total derivative: it gives the massless expression $2\sqrt{|g|}\,(C\gamma^\mu D_\mu\Phi)_A$, while $\Phi^T C\gamma^\mu\Phi = 0$ identically. The Dirac-type $\mathcal{L}$ with $\bar\Phi = \Phi^\dagger C$ is used for both fields: it is real, carries the mass term, and gives the same field equation for both statistics.

statement	**Wolfram checks (theory)**	**sympy checks (theory)**
$\mathcal{L}$ real	`L_real_C`	`commuting_lagran gian_real`, `commutin g_controls_not_vacuous`
total-divergence relation	`L_total_divergen ce_to_unsymmetrised_C`	`commuting_total_ divergence_relation`
spin connection drops out of $\mathcal{L}$, enters through the divergence identity; same equations as the connection-free symmetric Lagrangian	`L_spin_connectio n_drops_out_C`, `gammaOme ga_divergence_form`, `connection_free_ lagrangian_same_ equations` (scope)	`anticommutator_g amma_Omega_vanishes`, `divergence_of_sq rtg_gamma`, `connecti on_free_lagrangi an_same_equations` (scope)
negative control: Majorana-type L_g	`Majorana_Lg_tota l_derivative_grassmann`, `Majorana_Lg_comm uting_control`	`negative_control _majorana_grassm ann_total_derivative`, `negative_control _majorana_commut ing_contrast`

5 The exact field equations

Covariant form. The Euler-Lagrange equations of $\mathcal{L}$ for the commuting field (variation with respect to Φ^* and to Φ) are

$$\gamma^\mu D_\mu\Phi = \big(m + U'(S)\big)\,\Phi, \qquad (D_\mu\bar\Phi)\,\gamma^\mu = -\big(m + U'(S)\big)\,\bar\Phi,$$

and the second is the Dirac conjugate of the first. For $U = \frac{\lambda}{2}S^2$: $\gamma^\mu D_\mu\Phi = (m+\lambda S)\Phi$. (The sympy derivation includes the control that the same comparison with $m \to -m$ fails.)

Explicit form in the primordial field ($z = 6Hx_8$):

$$\begin{aligned} e^{-a_4}\sin^{-1/6} z\,\big(\gamma^{(x_1)}\partial_1 + \gamma^{(x_2)}\partial_2 + \gamma^{(x_3)}\partial_3\big)\Phi + \gamma^{(x_4)}\partial_4\Phi \\ + e^{a_4}\sin^{-1/6} z\,\big(\gamma^{(x_5)}\partial_5 + \gamma^{(x_6)}\partial_6 + \gamma^{(x_7)}\partial_7\big)\Phi \\ + \tan z\,\gamma^{(x_8)}\partial_8\Phi + 3H\,\gamma^{(x_8)}\Phi = \big(m + U'(S)\big)\,\Phi, \end{aligned}$$

and the adjoint equation

$$
\begin{aligned}
e^{-a_4}\sin^{-1/6} z\,\big(\partial_1\bar\Phi\,\gamma^{(x_1)}+\partial_2\bar\Phi\,\gamma^{(x_2)}+\partial_3\bar\Phi\,\gamma^{(x_3)}\big)+\partial_4\bar\Phi\,\gamma^{(x_4)}\\
+e^{a_4}\sin^{-1/6} z\,\big(\partial_5\bar\Phi\,\gamma^{(x_5)}+\partial_6\bar\Phi\,\gamma^{(x_6)}+\partial_7\bar\Phi\,\gamma^{(x_7)}\big)\\
+\tan z\,\partial_8\bar\Phi\,\gamma^{(x_8)}+3H\,\bar\Phi\,\gamma^{(x_8)}=-\big(m+U'(S)\big)\,\bar\Phi.
\end{aligned}
$$

The derivative coefficients are $1/f_a$: the inverse 3-space scale factor $e^{-a_4}\sin^{-1/6} z$, the inverse extra-time scale factor $e^{a_4}\sin^{-1/6} z$ (it grows as the extra times deflate) and $\tan z$ for the hidden direction; the last term on the left is the spin-connection term.

Evolution form. Since $(\gamma^{(x_4)})^2=-1$, the field equation is equivalent to the first-order evolution equation

$$
\partial_4\Phi=-\gamma^{(x_4)}\Big[\big(m+U'(S)\big)\Phi-\sum_{a\neq x_4}\frac{1}{f_a}\,\gamma^{(a)}\partial_a\Phi-3H\,\gamma^{(x_8)}\Phi\Big];
$$

the slices $x_4=\text{const}$ are non-characteristic ($g^{44}=-1\neq 0$).

Chiral block form. With $\Phi=(\varphi_-,\varphi_+)$ ($\Gamma=-1$ on rows 1-8, $\Gamma=+1$ on rows 9-16) every gamma is block off-diagonal, with 8×8 blocks $\bar\tau_a$ (upper right) and τ_a (lower left) listed in `Revision/theory/field-theory.json` (`field_equation_blocks`), and the field equation is the pair

$$
\sum_a\frac{1}{f_a}\,\bar\tau_a\,\partial_a\varphi_+ + 3H\,\bar\tau_{x_8}\varphi_+ = V\varphi_-,\qquad \sum_a\frac{1}{f_a}\,\tau_a\,\partial_a\varphi_- + 3H\,\tau_{x_8}\varphi_- = V\varphi_+.
$$

The mass term and the gravitational term $3H$ both couple the two chiral halves.

The 16 component equations. With $u=e^{-a_4}\sin^{-1/6} z$, $v=e^{a_4}\sin^{-1/6} z$, $t=\tan z$, $V=m+U'(S)$ and Φ_A the components in the row order of `Revision/algebra/gammas.json`, the equation $(\gamma^\mu D_\mu\Phi)_A=V\Phi_A$ reads (generated from `field_equation_components` of `Revision/theory/field-theory.json`, which the sympy record compares term by term):

$$
\begin{aligned}
(1)\quad &-\partial_4\Phi_{14}+u\,(-\partial_1\Phi_{16}+\partial_2\Phi_{15}-\partial_3\Phi_{14})\\
&+v\,(\partial_5\Phi_{15}+\partial_6\Phi_{16}+\partial_7\Phi_9)+(t\,\partial_8+3H)\,\Phi_9=V\,\Phi_1\\
(2)\quad &\partial_4\Phi_{13}+u\,(-\partial_1\Phi_{15}-\partial_2\Phi_{16}+\partial_3\Phi_{13})\\
&+v\,(\partial_5\Phi_{16}-\partial_6\Phi_{15}+\partial_7\Phi_{10})+(t\,\partial_8+3H)\,\Phi_{10}=V\,\Phi_2\\
(3)\quad &\partial_4\Phi_{16}+u\,(\partial_1\Phi_{14}-\partial_2\Phi_{13}-\partial_3\Phi_{16})\\
&+v\,(-\partial_5\Phi_{13}+\partial_6\Phi_{14}+\partial_7\Phi_{11})+(t\,\partial_8+3H)\,\Phi_{11}=V\,\Phi_3\\
(4)\quad &-\partial_4\Phi_{15}+u\,(\partial_1\Phi_{13}+\partial_2\Phi_{14}+\partial_3\Phi_{15})\\
&+v\,(-\partial_5\Phi_{14}-\partial_6\Phi_{13}+\partial_7\Phi_{12})+(t\,\partial_8+3H)\,\Phi_{12}=V\,\Phi_4
\end{aligned}
$$

$$
\begin{aligned}
(5)\;& \partial_4\Phi_{10} + u\,(-\partial_1\Phi_{12} + \partial_2\Phi_{11} - \partial_3\Phi_{10}) \\
&\quad + v\,(-\partial_5\Phi_{11} - \partial_6\Phi_{12} - \partial_7\Phi_{13}) + (t\,\partial_8 + 3H)\,\Phi_{13} = V\,\Phi_5 \\
(6)\;& -\partial_4\Phi_9 + u\,(-\partial_1\Phi_{11} - \partial_2\Phi_{12} + \partial_3\Phi_9) \\
&\quad + v\,(-\partial_5\Phi_{12} + \partial_6\Phi_{11} - \partial_7\Phi_{14}) + (t\,\partial_8 + 3H)\,\Phi_{14} = V\,\Phi_6 \\
(7)\;& -\partial_4\Phi_{12} + u\,(\partial_1\Phi_{10} - \partial_2\Phi_9 - \partial_3\Phi_{12}) \\
&\quad + v\,(\partial_5\Phi_9 - \partial_6\Phi_{10} - \partial_7\Phi_{15}) + (t\,\partial_8 + 3H)\,\Phi_{15} = V\,\Phi_7 \\
(8)\;& \partial_4\Phi_{11} + u\,(\partial_1\Phi_9 + \partial_2\Phi_{10} + \partial_3\Phi_{11}) \\
&\quad + v\,(\partial_5\Phi_{10} + \partial_6\Phi_9 - \partial_7\Phi_{16}) + (t\,\partial_8 + 3H)\,\Phi_{16} = V\,\Phi_8
\end{aligned}
$$

$$
\begin{aligned}
(9)\;& \partial_4\Phi_6 + u\,(\partial_1\Phi_8 - \partial_2\Phi_7 + \partial_3\Phi_6) \\
&\quad + v\,(-\partial_5\Phi_7 - \partial_6\Phi_8 - \partial_7\Phi_1) + (t\,\partial_8 + 3H)\,\Phi_1 = V\,\Phi_9 \\
(10)\;& -\partial_4\Phi_5 + u\,(\partial_1\Phi_7 + \partial_2\Phi_8 - \partial_3\Phi_5) \\
&\quad + v\,(-\partial_5\Phi_8 + \partial_6\Phi_7 - \partial_7\Phi_2) + (t\,\partial_8 + 3H)\,\Phi_2 = V\,\Phi_{10} \\
(11)\;& -\partial_4\Phi_8 + u\,(-\partial_1\Phi_6 + \partial_2\Phi_5 + \partial_3\Phi_8) \\
&\quad + v\,(\partial_5\Phi_5 - \partial_6\Phi_6 - \partial_7\Phi_3) + (t\,\partial_8 + 3H)\,\Phi_3 = V\,\Phi_{11} \\
(12)\;& \partial_4\Phi_7 + u\,(-\partial_1\Phi_5 - \partial_2\Phi_6 - \partial_3\Phi_7) \\
&\quad + v\,(\partial_5\Phi_6 + \partial_6\Phi_5 - \partial_7\Phi_4) + (t\,\partial_8 + 3H)\,\Phi_4 = V\,\Phi_{12}
\end{aligned}
$$

$$
\begin{aligned}
(13)\;& -\partial_4\Phi_2 + u\,(\partial_1\Phi_4 - \partial_2\Phi_3 + \partial_3\Phi_2) \\
&\quad + v\,(\partial_5\Phi_3 + \partial_6\Phi_4 + \partial_7\Phi_5) + (t\,\partial_8 + 3H)\,\Phi_5 = V\,\Phi_{13} \\
(14)\;& \partial_4\Phi_1 + u\,(\partial_1\Phi_3 + \partial_2\Phi_4 - \partial_3\Phi_1) \\
&\quad + v\,(\partial_5\Phi_4 - \partial_6\Phi_3 + \partial_7\Phi_6) + (t\,\partial_8 + 3H)\,\Phi_6 = V\,\Phi_{14} \\
(15)\;& \partial_4\Phi_4 + u\,(-\partial_1\Phi_2 + \partial_2\Phi_1 + \partial_3\Phi_4) \\
&\quad + v\,(-\partial_5\Phi_1 + \partial_6\Phi_2 + \partial_7\Phi_7) + (t\,\partial_8 + 3H)\,\Phi_7 = V\,\Phi_{15} \\
(16)\;& -\partial_4\Phi_3 + u\,(-\partial_1\Phi_1 - \partial_2\Phi_2 - \partial_3\Phi_3) \\
&\quad + v\,(-\partial_5\Phi_2 - \partial_6\Phi_1 + \partial_7\Phi_8) + (t\,\partial_8 + 3H)\,\Phi_8 = V\,\Phi_{16}
\end{aligned}
$$

In every component the gravitational term $3H\,\Phi_B$ appears next to the hidden-direction derivative $\tan z\,\partial_8\Phi_B$ of the same component B ($\gamma^{(x_8)}$ is a permutation matrix with entries $+1$).

Exact solutions used as test points. They satisfy the field equation for every a_4.

- $U = 0$: $\Phi = \sin^\alpha z\,\big(\cosh(kx_4) + \frac{\sinh(kx_4)}{k}M\big)\chi$ with constant χ (16 free complex constants), $M = -m\gamma^{(x_4)} + 3H(2\alpha+1)\gamma^{(x_4)}\gamma^{(x_8)}$, $k^2 = 9H^2(2\alpha+1)^2 - m^2$ and α arbitrary; $M^2 = k^2$ and $M^TC + CM = 0$. Honest scope: at these solutions the pressures and the momentum density vanish and $\rho = mS$ is constant in x_4, so they test the terms $\tan z\,\gamma^{(x_8)}\partial_8$ and $3H\gamma^{(x_8)}$ but are a weak test of conservation; the general proof of conservation is the Noether identity of section 7.
- $U = \frac{\lambda}{2}S^2$, homogeneous (nonlinear): $\Phi = \big(\cosh(kx_4) + \frac{\sinh(kx_4)}{k}M\big)\chi$, $M = -(m+\lambda S_0)\gamma^{(x_4)} + 3H\gamma^{(x_4)}\gamma^{(x_8)}$, $S_0 = \chi^\dagger C\chi$, $k^2 = 9H^2 - (m+\lambda S_0)^2$: $S = S_0$ is constant, the field equation holds, $\nabla_\mu T^\mu{}_\nu = 0$ for all ν, and $\rho = mS_0 + \lambda S_0^2/2$, $p = \lambda S_0^2/2$.

statement	Wolfram checks (theory)	sympy checks (theory)
Euler-Lagrange equations derived from $\mathcal{L}$	`EL_Psibar_C`, `EL_Psi_C`	`commuting_euler_lagrange_psibar_variation`, `commuting_euler_lagrange_psi_variation`
the adjoint equation is the Dirac conjugate	`adjoint_equation_is_Dirac_conjugate_C`	`commuting_adjoint_equation_is_conjugate`
explicit form, 16 components, chiral blocks	`Dirac_operator_explicit_C`, `block_form`	`commuting_euler_lagrange_psibar_variation` (the sympy field equation), compared term by term with `field_equation_components` and entry by entry with `field_equation_blocks` in the regenerated comparison record (section `comparison_with_wolfram` of the sympy report: 32 of 32 formula records agree)
evolution form	`evolution_form_C`	
exact solutions	`solution_matrix_square`, `exact_solution_x4_x8_C`, `exact_solution_nonlinear_homogeneous_C`	`exact_solution_family_x4_x8`, `exact_nonlinear_homogeneous_solution`

6 Non-triviality [2]: theorem and proof

The author's requirement [2]: the Lagrangian of dirac16complex00 must be non-trivial, i.e. its Euler-Lagrange equations always possess nonzero contributions from gravity through the canonical spin connection unless the spacetime is flat 4+4, and they must be self-consistent, checked and verified.

Theorem (NON-TRIVIALITY [2]). Hypotheses: the author's metric of section 1 with $H > 0$ and $0 < z < \pi/2$; a_4 any differentiable function of x_4 (constant included); the diagonal vielbein of section 3 and its canonical spin connection; Φ the commuting 16-component field with the Lagrangian of section 4 and any real potential U. Then statements (1) to (4) hold. Statements (1) and (2) concern the diagonal vielbein and the field variables Φ; statements (3) and (4) are the frame-independent content (see "Scope of [2]" below).

(1) The gravitational term of the field equation. The spin-connection part of the Euler-Lagrange operator is

$$\gamma^\mu D_\mu \Phi - \gamma^\mu \partial_\mu \Phi = \gamma^\mu \Omega_\mu \, \Phi, \qquad \gamma^\mu \Omega_\mu = 3H \, \gamma^{(x_8)},$$

a constant matrix, independent of a_4, x_4 and x_8. The term $3H\gamma^{(x_8)}\Phi$ is nonzero for every $H > 0$ and every $\Phi \neq 0$, because $\gamma^{(x_8)}$ is invertible, $(\gamma^{(x_8)})^2 = 1$.

(2) What cancels and what survives. Per direction (no sum)

$$\gamma^{x_i}\Omega_{x_i} = \tfrac{a_4'}{2}\,\gamma^{(x_4)} + \tfrac{H}{2}\,\gamma^{(x_8)} \qquad (i = x_1, x_2, x_3),$$
$$\gamma^{x_t}\Omega_{x_t} = -\tfrac{a_4'}{2}\,\gamma^{(x_4)} + \tfrac{H}{2}\,\gamma^{(x_8)} \qquad (t = x_5, x_6, x_7),$$

and $\gamma^{x_4}\Omega_{x_4} = \gamma^{x_8}\Omega_{x_8} = 0$. The time-direction ($\gamma^{(x_4)}$) terms of the 3 inflating and the 3 deflating directions cancel exactly, $3a_4'/2 - 3a_4'/2 = 0$; the hidden-direction ($\gamma^{(x_8)}$) terms of all six directions add, $6 \cdot H/2 = 3H$.

(3) It vanishes only in flat 4+4 space. $\Omega_\mu = 0$ for all μ if and only if $a_4' = 0$ and $H = 0$; $H = 0$ is a degenerate limit outside the family, so for every admissible metric the spin connection is nonzero. No choice of frame removes it: $[D_\mu, D_\nu] = \frac{1}{2}R_{ab\mu\nu}S^{ab}$ and $R^{x_8}{}_{x_8} = -6H^2 \neq 0$.

(4) The vielbein in every derivative term. Gravity also enters every derivative term,

$$\gamma^\mu\partial_\mu = e^{-a_4}\sin^{-1/6} z\,\gamma^{(x_i)}\partial_i + \gamma^{(x_4)}\partial_4 + e^{a_4}\sin^{-1/6} z\,\gamma^{(x_t)}\partial_t + \tan z\,\gamma^{(x_8)}\partial_8$$

(sums over i and t), none of the factors being identically 1; a_4 enters the field equation only through these factors (its spin-connection terms cancel by (2)).

Proof. Step (a), the inflating directions $i = x_1, x_2, x_3$: by section 3

$$\gamma^{x_i} = e^{-a_4}\sin^{-1/6} z\,\gamma^{(x_i)}, \qquad \Omega_{x_i} = \tfrac{1}{2}e^{a_4}\sin^{1/6} z\,\big(a_4'\gamma^{(x_i)}\gamma^{(x_4)} + H\gamma^{(x_i)}\gamma^{(x_8)}\big).$$

The scale factors cancel and $(\gamma^{(x_i)})^2 = \eta^{ii} = +1$, so $\gamma^{x_i}\Omega_{x_i} = \frac{1}{2}(a_4'\gamma^{(x_4)} + H\gamma^{(x_8)})$.

Step (b), the deflating extra times $t = x_5, x_6, x_7$:

$$\gamma^{x_t} = e^{a_4}\sin^{-1/6} z\,\gamma^{(x_t)}, \qquad \Omega_{x_t} = -\tfrac{1}{2}e^{-a_4}\sin^{1/6} z\,\big(a_4'\gamma^{(x_4)}\gamma^{(x_t)} + H\gamma^{(x_t)}\gamma^{(x_8)}\big).$$

With $\gamma^{(x_t)}\gamma^{(x_4)}\gamma^{(x_t)} = -\gamma^{(x_4)}(\gamma^{(x_t)})^2 = \gamma^{(x_4)}$ and $(\gamma^{(x_t)})^2 = -1$ this gives $\gamma^{x_t}\Omega_{x_t} = -\frac{1}{2}(a_4'\gamma^{(x_4)} - H\gamma^{(x_8)})$.

Step (c): $\Omega_{x_4} = \Omega_{x_8} = 0$. Summing (a), (b) and (c) gives statements (1) and (2). Equivalently, since $\{\gamma^\mu, \Omega_\mu\} = 0$ for each μ,

$$\gamma^\mu\Omega_\mu = \frac{1}{2\sqrt{|g|}}\,\partial_\mu(\sqrt{|g|}\,e_a{}^\mu)\gamma^{(a)};$$

only $\mu = x_8$ contributes: $\sqrt{|g|}\,e_{x_8}{}^{x_8} = \cos z\tan z = \sin z$ and $\frac{1}{2\cos z}\partial_8\sin z = 3H$, while $\partial_4(\sqrt{|g|}\,e_{x_4}{}^{x_4}) = \partial_4\cos z = 0$.

Step (d): each of the 12 nonzero $\omega_{\mu ab}$ is a_4' or H times $\pm e^{\pm a_4}\sin^{1/6} z \neq 0$, and the S^{ab} are linearly independent; this gives statement (3), with the curvature statement from the integrability identity and the Ricci components of section 3.

Step (e): statement (4) is the explicit form of section 5. Every step is verified exactly, for the author's T16 and, in the a_4 record, also for a second, independently built Clifford representation. □

What survives and what cancels, in words. The three inflating directions contribute $+\frac{a_4'}{2}\gamma^{(x_4)}$ each and the three deflating extra times $-\frac{a_4'}{2}\gamma^{(x_4)}$ each: the expansion-rate terms cancel exactly because 3-space inflates at the rate at which the extra times deflate (the same balance that makes $\sqrt{|g|} = \cos z$ independent of x_4). The warp $\sin^{1/6} z$ of the six directions along the hidden direction gives $+\frac{H}{2}\gamma^{(x_8)}$ for each of them and survives as $3H\gamma^{(x_8)}$.

Scope of [2] (exact). The value $\gamma^\mu\Omega_\mu = 3H\gamma^{(x_8)}$ belongs to the diagonal vielbein and to the field variables Φ:

- In another admissible vielbein of the same metric, the frame boosted in the (x_4, x_8) plane, $e'^{(x_4)} = \cosh b\, e^{(x_4)} + \sinh b\, e^{(x_8)}$, $e'^{(x_8)} = \sinh b\, e^{(x_4)} + \cosh b\, e^{(x_8)}$ with rapidity $b = \beta x_4 + b_0$, its canonical connection gives $\gamma'^\mu\Omega'_\mu = \frac{6H-\beta}{2}(\cosh b\,\gamma^{(x_8)} - \sinh b\,\gamma^{(x_4)})$, which vanishes identically for $\beta = 6H$, for every $H > 0$ and every a_4.
- Within the diagonal frame the rescaling $\Phi = \sin^{-1/2} z\,\chi$ removes the term exactly: $\gamma^\mu D_\mu\Phi = \sin^{-1/2} z\,\gamma^\mu\partial_\mu\chi$, so for $U = 0$ the equation for χ has no spin-connection term, and for $U = \frac{\lambda}{2}S^2$ the term becomes the x_8-dependent coupling $\lambda S[\chi]/\sin z$. The exact family of section 5 shows this: $\alpha = -\frac{1}{2}$ gives $M = -m\gamma^{(x_4)}$ with no H term.
- The canonical connection drops out of $\mathcal{L}$ (section 4), so the Euler-Lagrange equations are those of the connection-free symmetric Lagrangian; the deflation a_4 contributes nothing to $\gamma^\mu\Omega_\mu$ although $R^{x_4}{}_{x_4} = 6(a_4')^2 \neq 0$.
- What cannot be removed in any frame is Ω_μ itself (its curvature is the Riemann tensor, $R^{x_8}{}_{x_8} = -6H^2$), together with the frame factors $e^{\mp a_4}\sin^{-1/6} z$ and $\tan z$ of the derivative terms: this is the frame-independent content of [2]. In the diagonal frame the connection also enters the energy-momentum tensor (for a homogeneous configuration $K^{x_4}{}_{x_1}$ contains $\frac{1}{2}e^{a_4}\sin^{1/6} z\, H\,\bar\Phi\gamma^{(x_4)}\gamma^{(x_1)}\gamma^{(x_8)}\Phi$).
- Interpretation (labelled): in the quantum theory of dirac16complex the term makes the hidden-direction operator antisymmetric for the measure $\cos z\, dx_8$ in the variables Ψ, up to a boundary term that does not vanish at $z = \pi/2$; in the variables χ with the measure $dy = \cot z\, dx_8$ no H term is needed. This role therefore depends on the variables and the measure.

statement	**Wolfram checks**	**sympy checks**
(1) $\gamma^\mu\Omega_\mu = 3H\gamma^{(x_8)}$, nonzero for every $\Phi \neq 0$	`nontriviality_2_dirac16complex00`, `gammaOmega_equals_3H_gamma_x8` (theory); `gravity_term_gamma_mu_Omega_mu` (a_4 record)	`gamma_mu_Omega_mu_equals_3H_gamma_x8` (theory); `ownrep_gravity_term`, `authorT16_gravity_term` (a_4 record)
(2) the x_4 terms cancel, the x_8 terms add	`gammaOmega_x4_terms_cancel`, `gammaOmega_divergence_form`	`time_terms_cancel_hidden_term_survives`, `divergence_of_sqrtg_gamma`

statement	Wolfram checks	sympy checks
(3) $\Omega_\mu = 0$ only for $a_4' = 0$ and $H = 0$; never flat	`Omega_vanishes_iff_a4prime_and_H_vanish`, `never_flat_for_H_positive`, `spin_curvature_equals_Riemann`	`nontriviality_Omega_zero_iff_flat`, `curvature_nonzero_flat_only_formally`, `spinor_curvature_equals_riemann`
(4) explicit operator	`Dirac_operator_explicit_C`	
the same term in the quantum theory of dirac16complex (up to a boundary term)	`good_sector_hermiticity_curved`, `good_sector_hermiticity_up_to_the_brane_flux` (scope)	`good_sector_hermiticity_up_to_the_brane_flux` (scope)
scope: the boosted frame removes $\gamma^\mu\Omega_\mu$, not Ω_μ	`boosted_frame_reproduces_metric`, `boosted_frame_canonical_connection`, `boosted_frame_gammaOmega_formula`, `boosted_frame_gammaOmega_vanishes`, `boosted_frame_curvature_nonzero` (scope)	the same five names (scope)
scope: rescaling, deflation, energy-momentum tensor	`rescaling_removes_the_connection_term`, `rescaled_equation_quadratic_potential`, `gammaOmega_blind_to_the_deflation`, `spin_connection_in_the_energy_momentum_tensor` (scope)	the same four names (scope)

7 Self-consistency of the field equations

All statements are exact and verified for the commuting field.

Reality and derivation. $\mathcal{L}$ is real; the Euler-Lagrange equations derived from it are the field equation and its Dirac conjugate (sections 4 and 5).

Integrability of the connection. For all 28 pairs,

$$[D_\mu, D_\nu]\Phi = \big(\partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu]\big)\Phi = \tfrac{1}{2}R_{ab\mu\nu}S^{ab}\Phi.$$

Second-order consistency (Lichnerowicz identity). Identically for a general field,

$$(\gamma^\mu D_\mu)^2\Phi = g^{\mu\nu}\big(D_\mu D_\nu - \Gamma^\lambda{}_{\mu\nu}D_\lambda\big)\Phi - \frac{R}{4}\,\Phi, \qquad R = 6\big((a_4')^2 - 7H^2\big),$$

and $\gamma^\mu\gamma^\nu F_{\mu\nu} = -\frac{R}{2}$ for the spinor curvature $F_{\mu\nu} = [D_\mu, D_\nu]$.

Cauchy problem. The equation is first order in x_4 with non-characteristic slices (evolution form, section 5). This does not make the Cauchy problem well posed, and it is not listed here as a self-consistency property: for data that depend on the extra times the growth rates of the modes have no upper bound (with frozen coefficients $k_5 = K$ gives the rate $\sqrt{K^2 - m^2 - k_1^2 - k_2^2 - k_3^2 - k_8^2}$; Hadamard ill-posedness), and in the good sector, without a boundary condition at $z = \pi/2$, the x_8-independent modes grow whenever $m^2 < 9H^2$ (for $U = 0$; the matrix $-im\gamma^{(x_4)} + 3iH\gamma^{(x_4)}\gamma^{(x_8)}$ has square $(m^2 - 9H^2)I_{16}$).

Conserved current. $J^\mu = -i\bar\Phi\gamma^\mu\Phi$ is real, and off shell

$$\partial_\mu(\sqrt{|g|}\, J^\mu) = -i\sqrt{|g|}\,(\bar E\Phi + \bar\Phi E),$$

with $E = \gamma^\mu D_\mu\Phi - (m + U'(S))\Phi$ and $\bar E = (D_\mu\bar\Phi)\gamma^\mu + (m + U'(S))\bar\Phi$ (the convention of the Wolfram record), so $\nabla_\mu J^\mu = 0$ on shell. The charge density is $J^{x_4} = \Phi^\dagger B\Phi$, and the charge of a slice of constant x_4, $Q = \int \cos z\, \Phi^\dagger B\Phi\, d^7x$, is an indefinite form (B has signature (8,8)). Only the local law is established: Q is constant in x_4 only if no charge flows through the boundary of the slice, and at the brane $z = \pi/2$ the flow is not zero in general (`good_sector_hermiticity_up_to_the_brane_flux`, both engines; the lead's independent record `Revision/lead_checks/reports/charge-conjugation-and-u1.json` states the same for the current of dirac16complex). This record imposes no boundary condition at $z = \pi/2$ (section 15).

Bianchi-type identity of the energy-momentum tensor. The off-shell Noether identity of diffeomorphism invariance,

$$\partial_\mu(\sqrt{|g|}\, T^\mu{}_\lambda) - \sqrt{|g|}\, T^\mu{}_a\, \partial_\lambda e^a{}_\mu = \sum_\phi \mathrm{EL}_\phi\, \partial_\lambda\phi \qquad (\lambda = x_1, \ldots, x_8),$$

and the off-shell identity of local Lorentz invariance (the antisymmetric part of T is a combination of the Euler-Lagrange expressions EL_ϕ) give: on shell T is symmetric and $\nabla_\mu T^\mu{}_\nu = 0$, exactly, for general solutions.

Gravity side. Each Lovelock tensor is divergence-free, $\nabla_\mu E_{(k)}{}^\mu{}_\nu = 0$, and the constraint is propagated by the evolution equation (section 11).

Test points. The exact solutions of section 5 satisfy the field equation, current conservation and $\nabla_\mu T^\mu{}_\nu = 0$ exactly.

statement	**Wolfram checks (theory)**	**sympy checks (theory)**
integrability $[D_\mu, D_\nu] = \frac{1}{2}R_{ab\mu\nu}S^{ab}$	`spin_curvature_equals_Riemann`	`spinor_curvature_equals_riemann`
Lichnerowicz identity	`Lichnerowicz_identity_C`	`lichnerowicz_identity_on_fields`, `lichnerowicz_contraction`
current: reality, conservation, indefinite charge	`current_real_C`, `current_conservation_identity_C`, `charge_density_is_Krein_form_C`	`commuting_current_conservation`

statement	Wolfram checks (theory)	sympy checks (theory)
Noether identities and conservation of T	`Noether_identity_diffeomorphisms_C`, `Noether_identity_local_Lorentz_C`, `conservation_on_shell_general`	`commuting_emt_conservation_on_shell`, `commuting_emt_conservation_negative_control`
no well-posed Cauchy problem: unbounded growth, growing good-sector modes without a boundary condition	`extra_time_growth_rates_unbounded`, `good_sector_x8_independent_modes_without_boundary_condition` (scope)	the same two names (scope)

8 The energy-momentum tensor (classical bilinears)

For dirac16complex00 the energy-momentum tensor is the classical bilinear (a c-number function of Φ; no operator ordering, no normal ordering).

Definition and closed form. ${T^{\nu}}_{\mu} = {e^{b}}_{\mu}\,\frac{1}{\sqrt{|g|}}\,\delta S_{\mathrm{act}}/\delta {e^{b}}_{\nu}$ (exact first-order variation of $\sqrt{|g|}$, γ^{μ} and the spin connection; S_{act} the action) equals, for all 64 components,

$$ {T^{\nu}}_{\mu} = \delta^{\nu}_{\mu}\,L_0 - \tfrac{1}{2}(\bar{\Phi}\gamma^{\nu}D_{\mu}\Phi - D_{\mu}\bar{\Phi}\,\gamma^{\nu}\Phi) - \tfrac{1}{4}\,g_{\mu\rho}\nabla_{\lambda}(\bar{\Phi}\{\gamma^{\lambda},\Sigma^{\nu\rho}\}\Phi), \qquad \Sigma^{\nu\rho} = \tfrac{1}{4}[\gamma^{\nu},\gamma^{\rho}], $$

with $L_0 = \mathcal{L}/\sqrt{|g|}$; the last term (the totally antisymmetric spin density) comes from the variation of the spin connection. Its symmetric part is the Belinfante tensor

$$ {T^{\nu}}_{\mu} = \delta^{\nu}_{\mu}\,L_0 - \tfrac{1}{4}(\bar{\Phi}\gamma^{\nu}D_{\mu}\Phi - D_{\mu}\bar{\Phi}\,\gamma^{\nu}\Phi + \bar{\Phi}\gamma_{\mu}D^{\nu}\Phi - D^{\nu}\bar{\Phi}\,\gamma_{\mu}\Phi), $$

and on shell the variation tensor is symmetric and equals it (section 7; independently, with all 64 components of a first-order vielbein variation around the author's metric, the variation tensor equals the Belinfante tensor on shell for 64 of 64 components). The normalisation is $\delta S_{\mathrm{act}} = \frac{1}{2}\int \sqrt{|g|}\,T^{\mu\nu}\delta g_{\mu\nu}$.

Sign convention. The energy density is $\rho = -{T^{x_4}}_{x_4}$ and the pressures are $p_{\mu} = {T^{\mu}}_{\mu}$ (no sum) for the space-like and the extra-time directions:

$$ \rho = -{T^{x_4}}_{x_4}, \qquad p_3 = {T^{x_1}}_{x_1}, \qquad p_t = {T^{x_5}}_{x_5}, \qquad p_8 = {T^{x_8}}_{x_8}. $$

Diagonal components in the primordial field. With the kinetic term of direction μ (no sum)

$$ K_{\mu} = \frac{1}{2f_{\mu}}(\bar{\Phi}\gamma^{(\mu)}\partial_{\mu}\Phi - \partial_{\mu}\bar{\Phi}\,\gamma^{(\mu)}\Phi), $$

that is

$$K_{x_i} = \tfrac{1}{2} e^{-a_4} \sin^{-1/6} z \, (\bar{\Phi}\gamma^{(x_i)}\partial_i\Phi - \partial_i\bar{\Phi}\,\gamma^{(x_i)}\Phi), \qquad K_{x_4} = \tfrac{1}{2}(\bar{\Phi}\gamma^{(x_4)}\partial_4\Phi - \partial_4\bar{\Phi}\,\gamma^{(x_4)}\Phi),$$
$$K_{x_t} = \tfrac{1}{2} e^{a_4} \sin^{-1/6} z \, (\bar{\Phi}\gamma^{(x_t)}\partial_t\Phi - \partial_t\bar{\Phi}\,\gamma^{(x_t)}\Phi), \qquad K_{x_8} = \tfrac{1}{2} \tan z \, (\bar{\Phi}\gamma^{(x_8)}\partial_8\Phi - \partial_8\bar{\Phi}\,\gamma^{(x_8)}\Phi),$$

one has

$$L_0 = \sum_\mu K_\mu - mS - U(S), \qquad T^\mu{}_\mu = L_0 - K_\mu \ \text{(no sum)}.$$

No spin-connection term survives in the diagonal components ($\{\gamma^\mu, \Omega_\mu\} = 0$), and the variation tensor and the Belinfante tensor agree on the diagonal off shell. Hence, for every configuration (off shell),

$$\rho = -\sum_{\mu \neq x_4} K_\mu + mS + U(S),$$
$$p_3 = \sum_{\mu \neq x_1} K_\mu - mS - U(S) \quad (= T^{x_2}{}_{x_2} = T^{x_3}{}_{x_3} \text{ for states isotropic in 3-space}),$$
$$p_t = \sum_{\mu \neq x_5} K_\mu - mS - U(S), \qquad p_8 = \sum_{\mu \neq x_8} K_\mu - mS - U(S).$$

The x_4-x_8 component (the off-diagonal component allowed by the symmetry of the metric; $\Omega_{x_4} = \Omega_{x_8} = 0$):

$$T^{x_4}{}_{x_8} = -\tfrac{1}{4}(B_{48} - \cot z \, B_{84}), \qquad T^{x_8}{}_{x_4} = -\tan^2 z \, T^{x_4}{}_{x_8},$$
$$B_{48} = \bar{\Phi}\gamma^{(x_4)}\partial_8\Phi - \partial_8\bar{\Phi}\,\gamma^{(x_4)}\Phi, \qquad B_{84} = \bar{\Phi}\gamma^{(x_8)}\partial_4\Phi - \partial_4\bar{\Phi}\,\gamma^{(x_8)}\Phi;$$

On homogeneous on-shell states both $T^{x_4}{}_{x_8}$ and $T^{x_8}{}_{x_4}$ vanish identically, so these states do not source the x_4-x_8 component of the a_4 equations; this is not claimed for the other off-diagonal components (42 of the remaining 54 are nonzero bilinears in general), whose vanishing is a separate condition on the state (section 12).

Trace. Off shell (an exact identity in both engines) $T^\mu{}_\mu = 8L_0 - \sum_\mu K_\mu = 7\sum_\mu K_\mu - 8(mS + U)$. Exactly, $\sum_\mu K_\mu = (m + U')S + \frac{1}{2}(\bar{\Phi}E - \bar{E}\Phi)$, so on shell $\sum_\mu K_\mu = (m + U'(S))S$, $L_0 = S\,U'(S) - U(S)$ and

$$T^\mu{}_\mu = -mS + 7S\,U'(S) - 8U(S) = -mS + 3\lambda S^2 \quad (U = \tfrac{\lambda}{2}S^2).$$

Conservation and energy exchange. For a diagonal tensor

$$T = \mathrm{diag}(p_3, p_3, p_3, -\rho, p_t, p_t, p_t, p_8)(x_4, x_8)$$

whose entries are functions of x_4 and x_8, all eight components of the divergence are

$$\nabla_\mu T^\mu{}_{x_4} = -\partial_4\rho - 3a_4'(p_3 - p_t), \qquad \nabla_\mu T^\mu{}_{x_8} = \partial_8 p_8 + 3H \cot z \, (2p_8 - p_3 - p_t),$$

and $\nabla_\mu T^\mu{}_\nu = 0$ identically for $\nu = x_1, x_2, x_3, x_5, x_6, x_7$. Conservation is therefore exactly $\partial_4 \rho = -3a_4'(p_3 - p_t)$ and $\partial_8 p_8 = -3H \cot z\,(2p_8 - p_3 - p_t)$; for a homogeneous tensor (entries functions of x_4 only) it gives

$$\frac{d\rho}{dx_4} = -3\,a_4'\,(p_3 - p_t), \qquad p_3 + p_t = 2p_8 :$$

energy flows between 3-space and the extra times unless $p_3 = p_t$, since $\sqrt{|g|}$ does not depend on x_4.

statement	**Wolfram checks (theory)**	**sympy checks (theory)**
variation tensor in closed form	`T_vielbein_variation_closed_form_C`	`commuting_emt_equals_general_vielbein_variation_on_shell`
Belinfante tensor, symmetric	`T_symmetric_part_Belinfante_C`	`commuting_emt_symmetric`
normalisation $\delta S_{\rm act} = \frac{1}{2}\int \sqrt{\lvert g\rvert}\, T^{\mu\nu}\delta g_{\mu\nu}$, diagonal components	`T_diagonal_components_C`	`commuting_emt_equals_vielbein_variation_diagonal`
x_4-x_8 component	`T_x4_x8_component_C`	`commuting_T_x4x8_homogeneous`
trace, on-shell kinetic sum	`EMT_trace_C`, `kinetic_sum_on_shell_C`	`commuting_trace_on_shell`
energy exchange and the x_8 condition	`energy_exchange_equation`	`energy_exchange_equation`

9 Kinetic energy, potential energy, energy density and pressures

Split. $T^\nu{}_\mu = T_{\rm kin}{}^\nu{}_\mu + T_{\rm pot}{}^\nu{}_\mu$ with the potential part and the kinetic (derivative) part

$$T_{\rm pot}{}^\nu{}_\mu = -\delta^\nu_\mu\,(mS + U(S)),$$
$$T_{\rm kin}{}^\nu{}_\mu = \delta^\nu_\mu \sum_\lambda K_\lambda - \tfrac{1}{4}(\bar\Phi\gamma^\nu D_\mu\Phi - D_\mu\bar\Phi\,\gamma^\nu\Phi + \bar\Phi\gamma_\mu D^\nu\Phi - D^\nu\bar\Phi\,\gamma_\mu\Phi)$$

(formula `EMT_kinetic_potential` of `Revision/theory/field-theory.json`, compared with the sympy split).

Densities (per unit proper 7-volume; the energy in a region of a slice $x_4 = \text{const}$ is the integral with the weight $\sqrt{|g|} = \cos z$):

quantity	**kinetic part**	**potential part**	**total**
energy density $\rho = -T^{x_4}{}_{x_4}$	$-\sum_{\mu\neq x_4} K_\mu$	$mS + U(S)$	$-\sum_{\mu\neq x_4} K_\mu + mS + U$
3-space pressure $p_3 = T^{x_1}{}_{x_1}$	$\sum_{\mu\neq x_1} K_\mu$	$-(mS + U)$	$\sum_{\mu\neq x_1} K_\mu - mS - U$
extra-time pressure $p_t = T^{x_5}{}_{x_5}$	$\sum_{\mu\neq x_5} K_\mu$	$-(mS + U)$	$\sum_{\mu\neq x_5} K_\mu - mS - U$

quantity	kinetic part	potential part	total
hidden-direction pressure $p_8 = T^{x_8}{}_{x_8}$	$\sum_{\mu\neq x_8} K_\mu$	$-(mS+U)$	$\sum_{\mu\neq x_8} K_\mu - mS - U$

Homogeneous states on shell (Φ depending on x_4 only): $K_\mu = 0$ for $\mu \neq x_4$ and $K_{x_4} = (m + U'(S))S$. So the kinetic energy density vanishes, $-T_{\text{kin}}{}^{x_4}{}_{x_4} = 0$, the kinetic pressure is $(m + U'(S))S$ in every direction, and

$$\rho = mS + U(S), \qquad p_3 = p_t = p_8 = S\,U'(S) - U(S);$$

for $U = \frac{\lambda}{2}S^2$: $\rho = mS + \frac{\lambda}{2}S^2$ and $p = \frac{\lambda}{2}S^2$ (formula `EMT_homogeneous_on_shell` of `Revision /theory/field-theory.json`; the exact nonlinear solution of section 5 realises these values). S is constant along x_4 for a homogeneous solution, so ρ and p are constant, consistent with $d\rho/dx_4 = -3a_4'(p_3 - p_t) = 0$.

Sign of the energy (exact). The classical energy of dirac16complex00 is unbounded below already for $U = 0$ in the good sector. In the flat frame with $m = 2$ and $(k_1, k_2, k_3, k_8) = (1, 2, 0, 4)$, $E = 5$, the one-particle generator commutes with B and its $E = 5$ eigenspace (dimension 8) contains vectors u with $Bu = -u$ and with $Bu = +u$; the positive-frequency solutions $\Phi = c\,u\,e^{i(k\cdot x - 5x_4)}$ ($|u| = 1$) have the energy density $\rho = -T^{x_4}{}_{x_4} = -5|c|^2$ and $+5|c|^2$ and the charge density $\Phi^\dagger B\Phi = -|c|^2$ and $+|c|^2$ (`commuting_field_energy_unbounded_below`, both engines). By the Krein-inertia theorem of the pairing record (every real-frequency eigenspace has inertia (4,4), `Q_one_particle_Krein_inertia_real_frequencies`) such negative-energy modes exist at every real frequency. In the author's metric the homogeneous solutions of section 5 have $\rho = mS_0$ with $S_0 = \chi^\dagger C\chi$ of either sign. dirac16complex00 therefore has a negative-energy (ghost-like) sector at positive frequency, the classical form of the spin-statistics problem of a commuting field with a first-order Lagrangian; this matters for any reading of Hypothesis00, for example a phantom $w < -1$.

statement	Wolfram checks	sympy checks
homogeneous on-shell ρ, p; kinetic and potential parts	`exact_solution_nonlinear_homogeneous_C` (theory)	`commuting_homogeneous_on_shell_rho_p` (theory)
S constant along x_4 for a homogeneous solution	`condensate_S_constant` (a_4 record)	`ownrep_condensate_S_constant`, `authorT16_condensate_S_constant` (a_4 record)
energy unbounded below, indefinite charge	`commuting_field_energy_unbounded_below` (scope), `Q_one_particle_Krein_inertia_real_frequencies` (pairing)	`commuting_field_energy_unbounded_below` (scope), `Q.one_particle_Krein_inertia_proof` (pairing)

10 Equations of state

From the diagonal components (SPEC section 4):

$$w_3 = \frac{p_3}{\rho}, \qquad w_t = \frac{p_t}{\rho}, \qquad w_8 = \frac{p_8}{\rho}.$$

For homogeneous states on shell the three are equal,

$$w_3 = w_t = w_8 = \frac{S\,U'(S) - U(S)}{mS + U(S)}, \qquad w = \frac{\lambda S}{2m + \lambda S} \quad (U = \tfrac{\lambda}{2} S^2)$$

For $\lambda = 0$ the homogeneous state has $p = 0$ and $w = 0$. Because S, ρ and p are constant for a homogeneous solution, these ratios are constant in x_4.

statement	**source**
definitions and homogeneous values of w_3, w_t, w_8	formula `equation_of_state_definitions` of `Revision/theory/field-theory.json`; sympy check `commuting_homogeneous_on_shell_rho_p`

Not computed in this document: the equation of state that a 3-space observer infers (after integrating over the hidden direction and the extra times), its time dependence as the extra times deflate, CPL parameters, and any comparison with supernova data. These belong to the dark-sector investigation of Hypothesis00 (SPEC section 8), the separate Revision record `Revision/docs/DARK_SECTOR_HYPOTHESES`.

The verdict of the dark-sector record for dirac16complex00 (Hypothesis00; a summary, every number from its reports). The equation of state that a 3-space observer infers is not fixed by the field equations alone: the 4-dimensional density needs an ASSUMPTION about how the observer normalises over the deflating extra times, and the two normalisations of the dirac16complex00 record (N1, the scaling of the definitions (A) and (B) of that document, and N2, its definition (C)) differ by exactly -1 (`observer_N2_weff_identity`). Under (B) a positive-energy gas of good-sector modes has the time-varying, dark-matter-like law from 1/3 to 0, and a condensate is dust-like. Dark-energy-like values occur only under (C), as freezing evolution with $w \geq -1$ at every scale factor (M2, positive-energy good-sector modes without extra-time momentum, whose WKB frequencies $\omega^2 = m^2 + k^2/a^2$ stay real within the WKB model of the dark-sector record, an APPROXIMATION; in the author's metric, without a boundary condition at $z = \pi/2$, the x_8-independent good-sector modes grow whenever $m^2 < 9H^2$ (for $U = 0$), section 7) or as thawing evolution with $w \geq -1$ only before the turning point a_* of the extra-time mode (M3: $a_* = 1.8434$, M4: $a_* = 1.2355$); past a_* the mode grows, its frequency is imaginary and the bound is not established. The model M4 reproduces the Unite CPL tangent $(w_0, w_a) = (-0.861, -0.60)$ because its two free parameters were CHOSEN to solve the two tangent conditions, and its w does not cross -1 on $a \in [1/300, 1]$ (`M4_tangent_equals_unite`, `M4_never_phantom`); a crossing of $w = -1$ occurs only with a component of negative classical energy, a ghost-like sector (`M5_crosses_minus_1`, `M5_without_ghost_no_crossing`); the Unite constant $w = -0.764$ is the constant ratio of a condensate only for the tuned value $\lambda S/m = -382/441$ (`condensate_ratio_equals_unite_constant_w`). For a condensate that is the source of the metric in Einstein gravity (the linear member, section 12) the ratio is constant and below -1 for $\kappa\rho > 0$, above -1 for $\kappa\rho < 0$ (`linear_member_einstein_phantom_ratio`, whose detail states the hypotheses $\kappa > 0$ and $\rho > 0$; the sign of κ is not fixed by these records). The sign of $\kappa\rho$ is fixed by

the constraint: on the linear member $\kappa\rho = -(3A^2+21)H^2 - \Lambda$ (`Revision/field_equations_a4/a4-equations.json`, keys `linearMember/rhoEinstein` and `einstein/allowedA4`), so the ratio $w = -1 + 6(1+A^2)H^2/((3A^2+21)H^2+\Lambda)$ is phantom exactly when $\Lambda < -(21+3A^2)H^2$, and for $\Lambda = 0$ it is $w = (A^2-5)/(A^2+7) > -1$. The expansion seen by the observer gives $w = -1$ exactly (`expansion_inferred_w_einstein`). A model with as many tuned parameters as matched numbers reproduces them by construction and is not a prediction. That record establishes neither Hypothesis nor Hypothesis00; it states what each field can and cannot produce under its stated assumptions.

11 The field equations for a4[x4]

Einstein-Lovelock equations (SPEC section 5):

$$\sum_{k=1}^{3} \alpha_k \, E_{(k)}{}^{\mu}{}_{\nu} + \Lambda \, \delta^{\mu}_{\nu} = \kappa \, T^{\mu}{}_{\nu}, \qquad E_{(k)} = -\frac{P_{(k)}}{2^{k+1}}, \qquad E_{(1)} = G,$$

with $P_{(k)}{}^{h}{}_{j}$ the Lovelock tensors computed with the generalized Kronecker delta (GKD) from the Riemann tensor (MTW convention). $P_{(4)}$ vanishes in 8 dimensions (a GKD with 9 indices over 8 values), so $k = 1, 2, 3$ is the complete series. The components computed directly from the curvature equal the GKD record `Revision/gkd_lovelock/results/lovelock-tensors.json` monomial by monomial. The mixed Riemann components contain neither the warp nor e^{a_4}, so the left-hand sides depend on H, a_4' and a_4'' only.

The Lovelock components. All off-diagonal components vanish (including x_4-x_8); $x_2 = x_3 = x_1$ and $x_6 = x_7 = x_5$:

$$\begin{aligned}
E_{(1)}{}^{x_1}{}_{x_1} &= -3(a_4')^2 + a_4'' + 15H^2, \qquad E_{(1)}{}^{x_4}{}_{x_4} = 3(a_4')^2 + 21H^2,\\
E_{(1)}{}^{x_5}{}_{x_5} &= -3(a_4')^2 - a_4'' + 15H^2, \qquad E_{(1)}{}^{x_8}{}_{x_8} = 15H^2 - 3(a_4')^2,\\
E_{(2)}{}^{x_1}{}_{x_1} &= 12(a_4')^4 - 24(a_4')^2 a_4'' + 168(a_4')^2H^2 - 40a_4''H^2 - 180H^4,\\
E_{(2)}{}^{x_4}{}_{x_4} &= -36(a_4')^4 - 120(a_4')^2H^2 - 420H^4,\\
E_{(2)}{}^{x_5}{}_{x_5} &= 12(a_4')^4 + 24(a_4')^2 a_4'' + 168(a_4')^2H^2 + 40a_4''H^2 - 180H^4,\\
E_{(2)}{}^{x_8}{}_{x_8} &= 12(a_4')^4 + 168(a_4')^2H^2 - 180H^4,
\end{aligned}$$

$$\begin{aligned}
E_{(3)}{}^{x_1}{}_{x_1} &= -72(a_4')^6 + 360(a_4')^4 a_4'' - 648(a_4')^4H^2 + 432(a_4')^2 a_4''H^2\\
&\quad - 1944(a_4')^2H^4 + 360a_4''H^4 + 360H^6,\\
E_{(3)}{}^{x_4}{}_{x_4} &= 360(a_4')^6 + 648(a_4')^4H^2 + 1080(a_4')^2H^4 + 2520H^6,\\
E_{(3)}{}^{x_5}{}_{x_5} &= -72(a_4')^6 - 360(a_4')^4 a_4'' - 648(a_4')^4H^2 - 432(a_4')^2 a_4''H^2\\
&\quad - 1944(a_4')^2H^4 - 360a_4''H^4 + 360H^6,\\
E_{(3)}{}^{x_8}{}_{x_8} &= -72(a_4')^6 - 648(a_4')^4H^2 - 1944(a_4')^2H^4 + 360H^6.
\end{aligned}$$

Each $E_{(k)}$ is divergence-free, and $\sum_h P_{(k)}{}^{h}{}_{h} = (8-2k)L_{(k)}$.

The independent equations for a general source $T = \mathrm{diag}(p_3, p_3, p_3, -\rho, p_t, p_t, p_t, p_8)$ plus $q_{48} = T^{x_4}{}_{x_8}$ and $q_{84} = T^{x_8}{}_{x_4}$:

- constraint (x_4, first order in a_4): $\sum_k \alpha_k E_{(k)}{}^{x_4}{}_{x_4} + \Lambda = -\kappa\rho$;
- 3-space ($x_1 = x_2 = x_3$): $\sum_k \alpha_k E_{(k)}{}^{x_1}{}_{x_1} + \Lambda = \kappa p_3$;
- extra times ($x_5 = x_6 = x_7$): $\sum_k \alpha_k E_{(k)}{}^{x_5}{}_{x_5} + \Lambda = \kappa p_t$;
- hidden direction (x_8, first order in a_4): $\sum_k \alpha_k E_{(k)}{}^{x_8}{}_{x_8} + \Lambda = \kappa p_8$;
- off-diagonal: $0 = \kappa q_{48} = \kappa q_{84}$, and $0 = \kappa T^{\mu}{}_{\nu}$ for every other $\mu \neq \nu$.

The evolution equation (3-space minus extra time) contains a_4'' and is sourced by $p_3 - p_t$:

$$a_4'' \, F(a_4') = \kappa\,(p_3 - p_t), \qquad F = 720(a_4')^4\alpha_3 - 48(a_4')^2\alpha_2 + 864(a_4')^2\alpha_3 H^2 + 2\alpha_1 - 80\alpha_2 H^2 + 720\alpha_3 H^4;$$

F vanishes identically only for $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

Consistency conditions.

- Algebraic: $\sum_k \alpha_k (E^{x_1}{}_{x_1} + E^{x_5}{}_{x_5} - 2E^{x_8}{}_{x_8}) = 0$ identically, so the equations force $p_3 + p_t = 2p_8$.
- x_8-dependence: the left-hand sides are free of x_8 (and of a_4 itself), so every component of the source must be a function of x_4 alone and $q_{48} = q_{84} = 0$; an x_8-dependent source is not compatible with the metric of section 1.
- Isotropy: since the left-hand sides of x_1, x_2, x_3 coincide (and those of x_5, x_6, x_7), the source must have $T^{x_1}{}_{x_1} = T^{x_2}{}_{x_2} = T^{x_3}{}_{x_3}$ and $T^{x_5}{}_{x_5} = T^{x_6}{}_{x_6} = T^{x_7}{}_{x_7}$.
- Conservation: $\nabla_\mu T^{\mu}{}_{x_4} = -\partial_4\rho - 3a_4'(p_3 - p_t) + \partial_8 q_{84} - 6H\tan z\, q_{84}$ and $\nabla_\mu T^{\mu}{}_{x_8} = \partial_8 p_8 + 3H\cot z\,(2p_8 - p_3 - p_t) + \partial_4 q_{48}$; with an x_8-independent source and $q_{48} = q_{84} = 0$ they reduce to $\rho' = -3a_4'(p_3 - p_t)$ and $p_3 + p_t = 2p_8$, both implied by the field equations (Bianchi identity).
- Constraint propagation: $\frac{d}{dx_4}\sum_k \alpha_k E_{(k)}{}^{x_4}{}_{x_4} = 3a_4' \sum_k \alpha_k (E^{x_1}{}_{x_1} - E^{x_5}{}_{x_5})$ identically, so the derivative of the constraint is $3a_4'$ times the evolution equation once $\rho' = -3a_4'(p_3 - p_t)$ holds; for $a_4' \neq 0$ the evolution equation follows from the constraint and conservation.

Einstein gravity ($\alpha_1 = 1$, $\alpha_2 = \alpha_3 = 0$):

$$\begin{aligned}
&3(a_4')^2 + 21H^2 + \Lambda = -\kappa\rho, \qquad -3(a_4')^2 + a_4'' + 15H^2 + \Lambda = \kappa p_3,\\
&-3(a_4')^2 - a_4'' + 15H^2 + \Lambda = \kappa p_t, \qquad -3(a_4')^2 + 15H^2 + \Lambda = \kappa p_8,\\
&2a_4'' = \kappa(p_3 - p_t), \qquad \kappa(p_8 + \rho) = -6(a_4')^2 - 6H^2.
\end{aligned}$$

Every C^2 function a_4 is allowed with the source it requires; that source always has $\kappa(\rho + p_8) = -6((a_4')^2 + H^2) < 0$ (for $\kappa > 0$ it violates the null energy condition along $x_4 + x_8$), and $T = 0$ has no solution for $H > 0$ and any Λ.

The linear member $a_4 = AHx_4 + a_0$ ($a_4' = AH$, $a_4'' = 0$; 3-space scale factor e^{AHx_4}, extra-time scale factor e^{-AHx_4}: exponential deflation of the extra times for $A > 0$). For any α_k the field equations require $p_3 = p_t = p_8 = p$ and ρ, all constant, with

$$\begin{aligned}\rho = &-\frac{360A^6\alpha_3H^6}{\kappa} + \frac{36A^4\alpha_2H^4}{\kappa} - \frac{648A^4\alpha_3H^6}{\kappa} - \frac{3A^2\alpha_1H^2}{\kappa} + \frac{120A^2\alpha_2H^4}{\kappa}\\ &- \frac{1080A^2\alpha_3H^6}{\kappa} - \frac{21\alpha_1H^2}{\kappa} + \frac{420\alpha_2H^4}{\kappa} - \frac{2520\alpha_3H^6}{\kappa} - \frac{\Lambda}{\kappa},\\ p = &-\frac{72A^6\alpha_3H^6}{\kappa} + \frac{12A^4\alpha_2H^4}{\kappa} - \frac{648A^4\alpha_3H^6}{\kappa} - \frac{3A^2\alpha_1H^2}{\kappa} + \frac{168A^2\alpha_2H^4}{\kappa}\\ &- \frac{1944A^2\alpha_3H^6}{\kappa} + \frac{15\alpha_1H^2}{\kappa} - \frac{180\alpha_2H^4}{\kappa} + \frac{360\alpha_3H^6}{\kappa} + \frac{\Lambda}{\kappa};\end{aligned}$$

a vacuum ($\rho = p = 0$) needs

$$\mathcal{V} = 72A^4\alpha_3H^4 - 8A^2\alpha_2H^2 + 144A^2\alpha_3H^4 + \alpha_1 - 40\alpha_2H^2 + 360\alpha_3H^4 = 0,$$

which in Einstein gravity ($\mathcal{V} = 1$) is impossible. The equations contain A only through A^2: the deflating branch $A > 0$ (a_4 increasing) and the branch $A < 0$ satisfy the same equations.

statement	**Wolfram checks (a_4 record)**	**sympy checks (a_4 record)**
metric, volume, Riemann free of warp and e^{a_4}	`metric_is_SPEC_section_1`, `sqrt_abs_det_g_is_cos_z`, `mixed_riemann_free_of_warp_and_a4`	`mixed_riemann_free_of_warp_and_a4`, `riemann_entries_laurent`
$P_{(k)}$ from the curvature equal the GKD record; traces; $P_{(4)} = 0$	`P1_direct_equals_minus_4_Einstein`, `P1_direct_equals_gkd_branch_monomials`, `P2_direct_equals_gkd_branch_monomials`, `P3_direct_equals_gkd_branch_monomials`, `P1_trace_identity`, `P2_trace_identity`, `P3_trace_identity`, `P4_vanishes_pigeonhole`	`P1_equals_minus_4_Einstein`, `P1_equals_gkd_branch_monomials`, `P2_equals_gkd_branch_monomials`, `P3_equals_gkd_branch_monomials`
divergence-free $E_{(k)}$	`E1_divergence_free`, `E2_divergence_free`, `E3_divergence_free`	`E1_divergence_free`, `E2_divergence_free`, `E3_divergence_free`
independent components, structure	`independent_components`	`other_components_vanish`, `json_lovelock_components`, `json_general_source_equations`
evolution equation and F	`evolution_factorises_a4pp_times_F`, `evolution_F_not_identically_zero`	`evolution_factorises`, `evolution_F_not_identically_zero`

statement	**Wolfram checks (a_4 record)**	**sympy checks (a_4 record)**
algebraic condition, first-order constraint and x_8 equation	`algebraic_identity_x1_plus_x5_minus_2x8`, `x8_component_contains_no_a4pp`	`algebraic_identity`, `constraint_and_x8_first_order`
conservation, constraint propagation	`conservation_components`, `constraint_propagation_bianchi`	`conservation_components`, `bianchi_x4`
Einstein case	`einstein_components`, `einstein_null_energy_x8`, `einstein_no_vacuum_solution`	`einstein_components`, `einstein_null_energy_x8`, `einstein_no_vacuum`, `json_einstein`
linear member	`linear_member_equal_pressures`, `linear_member_vacuum_factor`	`linear_member_equal_pressures`, `linear_member_vacuum_factor`, `json_linear_member`

12 dirac16complex00 as the source of the a4 equations

The tensor. The source is the classical bilinear tensor of section 8. The a_4 record writes it as $T^{\mu}{}_{\nu} = \sigma_T\big(- K^{(\mu}{}_{\nu)} + \delta^{\mu}_{\nu}\, L_0\big)$ with $K^{\mu}{}_{\nu} = \frac{1}{2}(\bar{\Phi}\gamma^{\mu} D_{\nu}\Phi - D_{\nu}\bar{\Phi}\,\gamma^{\mu}\Phi)$ symmetrised, and $\sigma_T = +1$; this is the Belinfante tensor of section 8 (whose symmetrised kinetic term is the same expression), so the two records use the same tensor and the same sign. With $k_{\mu} = K^{\mu}{}_{\mu}$ (no sum; equal to K_{μ} of section 8) and $L_0 = \sum_{\mu} k_{\mu} - mS - U(S)$:

$$\rho = k_4 - L_0, \qquad p_3 = L_0 - k_1, \qquad p_t = L_0 - k_5, \qquad p_8 = L_0 - k_8, \qquad q_{48} = -K^{(x_4}{}_{x_8)}.$$

The equations with dirac16complex00 as the source (classical commuting Φ; k_{μ}, S and U classical bilinears):

$$\sum_k \alpha_k E_{(k)}{}^{x_4}{}_{x_4} + \Lambda = -\kappa\,(k_4 - L_0), \qquad \sum_k \alpha_k E_{(k)}{}^{x_8}{}_{x_8} + \Lambda = \kappa\,(L_0 - k_8),$$
$$a_4''\,F(a_4') = \kappa\,(k_5 - k_1), \qquad 2k_8 = k_1 + k_5, \qquad K^{(\mu}{}_{\nu)} = 0 \;\; (\mu \neq \nu),$$

together with $k_1 = k_2 = k_3$, $k_5 = k_6 = k_7$ and the requirement that every component be independent of x_8. On shell $L_0 = S\,U'(S) - U(S)$. The evolution equation shows what drives a_4'': a difference between the kinetic terms along 3-space and along the extra times. This record contains no exact non-homogeneous dirac16complex00 configuration that satisfies all of these conditions with $k_1 \neq k_5$; whether one exists is open.

The homogeneous condensate. For $\Phi = \Phi(x_4)$ on shell the field equation is $\partial_4\Phi = A_{\Phi}\Phi$ with $A_{\Phi} = -\gamma^{(x_4)}\big(M - 3H\gamma^{(x_8)}\big)$ and $M = m + U'(S)$, free of x_8. Proved exactly (Wolfram, and in the sympy record both for an independently built Clifford representation and for the author's T16):

- $CA_{\Phi} + A_{\Phi}^{T}C = 0$, so $S = \bar{\Phi}\Phi$ is constant along x_4;

- the diagonal kinetic components are ${K^{x_4}}_{x_4} = MS$ and 0 for every other direction, and ${K^{x_4}}_{x_8} = {K^{x_8}}_{x_4} = 0$; hence $\rho = mS + U(S)$, $p_3 = p_t = p_8 = S\,U'(S) - U(S)$ and $q_{48} = q_{84} = 0$, and the algebraic condition and the x_8-independence hold;
- the nonzero off-diagonal kinetic components are exactly multiples of the 15 three-gamma bilinears $B_{abc} = \bar{\Phi}\gamma^{(a)}\gamma^{(b)}\gamma^{(c)}\Phi$ with $\{a,b,c\} = \{i, x_4, x_8\}$ ($i = x_1..x_3, x_5..x_7$) and $\{i, j, x_4\}$ ($i = x_1..x_3$, $j = x_5..x_7$); there are 42 nonzero ordered pairs (μ, ν).

The 42 components fall into ten groups; within each group the coefficient is the same for every 3-space index x_i ($i = 1, 2, 3$) and extra-time index x_t ($t = 5, 6, 7$), and on the patch $\sqrt[12]{\cot^2 z + 1} = \sin^{-1/6} z$ (the gammas in this table are the frame gammas; generated from `Revision/field_equations_a4/a4-equations.json`):

component	bilinear	coefficient
${K^{x_i}}_{x_4}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$\frac{7}{4}e^{-a_4}\,\sqrt[12]{\cot^2 z + 1}\,H$
${K^{x_i}}_{x_t}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_t}\Phi$	$-\frac{1}{2}e^{-2a_4}(a_4')$
${K^{x_i}}_{x_8}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$\frac{1}{4}e^{-a_4}(a_4')\cot z\,\sqrt[12]{\cot^2 z + 1}$
${K^{x_4}}_{x_i}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$-\frac{7e^{a_4}H}{4\sqrt[12]{\cot^2 z+1}}$
${K^{x_4}}_{x_t}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$-\frac{7e^{-a_4}H}{4\sqrt[12]{\cot^2 z+1}}$
${K^{x_t}}_{x_i}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_t}\Phi$	$\frac{1}{2}e^{2a_4}(a_4')$
${K^{x_t}}_{x_4}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$-\frac{7}{4}e^{a_4}\,\sqrt[12]{\cot^2 z + 1}\,H$
${K^{x_t}}_{x_8}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$\frac{1}{4}e^{a_4}(a_4')\cot z\,\sqrt[12]{\cot^2 z + 1}$
${K^{x_8}}_{x_i}$	$\bar{\Phi}\gamma^{x_i}\gamma^{x_4}\gamma^{x_8}\Phi$	$\frac{e^{a_4}(a_4')}{4\cot z\,\sqrt[12]{\cot^2 z+1}}$
${K^{x_8}}_{x_t}$	$\bar{\Phi}\gamma^{x_4}\gamma^{x_t}\gamma^{x_8}\Phi$	$-\frac{e^{-a_4}(a_4')}{4\cot z\,\sqrt[12]{\cot^2 z+1}}$

The off-diagonal field equations $0 = \kappa T^{\mu}{}_{\nu}$ therefore require $B_{i\,x_4 x_8} = 0$ for the six i (coefficients proportional to H and to a_4') and, when $a_4' \neq 0$, $B_{ijx_4} = 0$ for the nine pairs i, j.

Exact witnesses that these conditions can be met with $S \neq 0$: $\Phi = e^{-i\omega x_4}\Phi_0$ with $\omega = \sqrt{M^2 - 9H^2}$ ($M^2 > 9H^2$), and $\Phi_0 = v_1 + c\,v_2$ built from the two eigenvectors of A_Φ in the sectors $\gamma^{(x_1)}\gamma^{(x_5)} = \gamma^{(x_2)}\gamma^{(x_6)} = \gamma^{(x_3)}\gamma^{(x_7)} = -1$ and $+1$, $c = \overline{v_1^\dagger C v_2}$. At $(M, H) = (5, 1)$, $(5, 4/3)$ and $(-5, 1)$ (exact Gaussian-rational arithmetic) the frequencies are $\omega = 4, 3, 4$, all 15 three-gamma bilinears and every off-diagonal kinetic component vanish for every a_4 and a_4', and $S = 51200$, 28800, 51200 respectively. These values are computed in the author's matrices T16; S scales with the arbitrary normalisation of the exact null vectors v_1, v_2 (in the comparison basis of the record, $\mathrm{Cl}(1,1)^{\otimes 4}$, the same witnesses give 204800, 115200, 204800), whereas ω, $S \neq 0$ and the vanishing of the bilinears do not depend on that normalisation.

Theorem (the condensate allows only the linear member). Hypotheses: Φ a homogeneous on-shell condensate as above (classical, commuting) satisfying the off-diagonal conditions; $a_4 \in C^2$; $(\alpha_1, \alpha_2, \alpha_3) \neq (0, 0, 0)$. Then $p_3 = p_t$, so $a_4'' F(a_4') = 0$; F is a nonzero polynomial in a_4' with finitely many roots and a_4' is continuous, so $a_4'' = 0$ on every interval:

$$a_4 = AHx_4 + a_0,$$

with real A: only the linear member is allowed (`theoremLinear` of `Revision/field_equations_a4/a4-equations.json`, resting on the checks listed below). The equations contain A only through

A^2 (invariant under $A \to -A$): the extra times deflate for $A > 0$, which is a choice of sign and is not selected by the equations; $A < 0$ is allowed on the same footing, and $A = 0$ is the static case.

The source it requires. With $a_4' = AH$ the remaining equations are the constraint and the x_8 equation, at $a_4' = AH$:

$$\kappa\,(mS + U) = -\Big(\Lambda + \sum_k \alpha_k {E_{(k)}}^{x_4}{}_{x_4}\Big), \qquad \kappa\,(S\,U' - U) = \Lambda + \sum_k \alpha_k {E_{(k)}}^{x_8}{}_{x_8},$$

that is, $\rho = mS + U$ and $p = SU' - U$ must equal the linear-member values of section 11. In Einstein gravity

$$\kappa\,(mS + U) = -(3A^2 + 21)H^2 - \Lambda, \qquad \kappa\,(S\,U' - U) = (15 - 3A^2)H^2 + \Lambda,$$

and for $U = \frac{\lambda}{2}S^2$ and $m \neq 0$ these are equivalent to

$$\kappa\, mS = -(36H^2 + 2\Lambda), \qquad 6(A^2 + 1)H^2 = -\kappa\, S\,(m + \lambda S),$$

so $A^2 = -\kappa S(m + \lambda S)/(6H^2) - 1$, and a real A needs $\kappa S(m + \lambda S) \leq -6H^2$. For the condensate $m + \lambda S = M$ is the effective mass of the field equation and $S(m + \lambda S) = \rho + p$; the requirement is the null-energy statement of section 11 for this source. S takes both signs (section 2), so the sign condition is not excluded by the algebra; the three exact witnesses above have $S > 0$ with $M = 5, 5, -5$. This record contains no check that combines a witness with the Einstein conditions (definite κ, Λ, m, λ and A): that the deflating member is realised by a specific dirac16complex00 condensate is not established here.

statement	**Wolfram checks (a_4 record)**	**sympy checks (a_4 record)**
the Clifford facts used, in the author's T16 (with the Cl(1,1)$^{\otimes 4}$ basis as an exactly equivalent comparison)	`clifford_relations_author_T16`, `C_properties_author_T16`, `gamma_mu_anticommutes_with_Omega_mu_no_sum`, `representations_equivalent_author_T16_own_rep`	`ownrep_clifford`, `ownrep_C_properties`, `ownrep_anticommutator_no_sum`, `authorT16_clifford`, `authorT16_C_properties`, `authorT16_anticommutator_no_sum`
condensate equation free of x_8; S constant; adjoint	`condensate_equation_x8_consistent`, `condensate_S_constant`, `condensate_adjoint_equation`	`ownrep_condensate_S_constant`, `ownrep_condensate_adjoint`, `authorT16_condensate_S_constant`, `authorT16_condensate_adjoint`
diagonal kinetic tensor of the condensate	`condensate_kinetic_tensor_diagonal`	`ownrep_condensate_kinetic_diagonal`, `authorT16_condensate_kinetic_diagonal`

statement	Wolfram checks (a_4 record)	sympy checks (a_4 record)
off-diagonal components: the 15 three-gamma bilinears, coefficients	`condensate_offdiagonal_are_three_gamma_bilinears`	`ownrep_condensate_offdiagonal_three_gamma`, `authorT16_condensate_offdiagonal_three_gamma`, `json_offdiagonal_coefficients`, `offdiagonal_coefficients_representation_independent`
exact witnesses	`condensate_diagonal_witness_exact`	`ownrep_condensate_witness`, `authorT16_condensate_witness`
the linear member is the only one ($F \neq 0$)	`evolution_F_not_identically_zero`, `linear_member_equal_pressures`	`evolution_F_not_identically_zero`, `linear_member_equal_pressures`
Einstein conditions with $U = \frac{\lambda}{2}S^2$	`condensate_einstein_quadratic_U`	`condensate_einstein_quadratic_U`

13 Pairing theorems for dirac16complex00 (summary)

The pairing of universes of masses $\{+m, -m\}$ is the subject of `Revision/docs/PAIR_CREATION_PROOFS`; for the commuting field the Revision pairing records prove exactly the following.

T1 (chirality pairing). Hypotheses: any gravitational field, fixed and the same for both members (a test field; the metric is not varied); $\Gamma = \mathrm{diag}(-I_8, I_8)$; $U = \frac{\lambda}{2}S^2$, or any function U with $U \to -U$. Then, identically,

$$\mathcal{L}_{m,\lambda}[\Gamma\Phi] = -\mathcal{L}_{-m,-\lambda}[\Phi], \qquad S[\Gamma\Phi] = S[\Phi], \qquad T^{(-m,-\lambda)}[\Gamma\Phi] = -T^{(m,\lambda)}[\Phi], \qquad J[\Gamma\Phi] = -J[\Phi]$$

(for a general U: $\mathcal{L}_{m,U}[\Gamma\Phi] = -\mathcal{L}_{-m,-U}[\Phi]$), and Φ solves the (m, λ) equations if and only if $\Gamma\Phi$ solves the $(-m, -\lambda)$ ones. The pair's total energy-momentum density, current and charge vanish as classical c-number bilinears.

T2 (mirror pairing). With the isometry $\phi : x_8 \to \pi/(6H) - x_8$ ($z \to \pi - z$, the Z_2 mirror across $z = \pi/2$, an ASSUMED construction at a degenerate surface of the metric) and $\Phi'(\phi(x)) = \gamma^{(x_8)}\Phi(x)$:

$$\mathcal{L}_{m,\lambda}[\Phi'](\phi(x)) = \mathcal{L}_{-m,\lambda}[\Phi](x), \qquad S' = -S,$$

and the mirror partner carries the same energy density and charge (the tensor transformed by the reflection), not the opposite.

There is no quantum reading for dirac16complex00: it is not quantised. These theorems map solutions to solutions; they derive no creation process, rate or amplitude (the lists of what is not established in the pairing records).

statement	Wolfram checks (pairing)	sympy checks (pairing)
T1 for the commuting field in the primordial metric	Gamma_properties, T1_kernel_field_equation, T1_linearity_argument, T1_Lagrangian_primordial_commuting, T1_Lagrangian_negative_controls_primordial_commuting, T1_field_equation_covariance_primordial_commuting, T1_Euler_Lagrange_map_primordial_commuting, T1_energy_momentum_primordial_commuting, T1_current_primordial_commuting, T1_general_potential_primordial_commuting	T1.metric.commuting.lagrangian, T1.metric.commuting.negative_controls, T1.metric.commuting.euler_lagrange_map, T1.metric.commuting.emt, T1.metric.commuting.pair_total_emt_zero, T1.metric.commuting.current, T1.metric.commuting.general_potential
T2 for the commuting field	T2_mirror_is_isometry, T2_mirror_Lagrangian_commuting, T2_mirror_energy_momentum_and_current_commuting, T2_mirror_Euler_Lagrange_commuting	T2.metric.commuting.euler_lagrange_map, T2.metric.commuting.emt, T2.metric.commuting.current, T2.metric.commuting.S_odd

14 Check index

Every check below has the verdict PASS in the named report (the publication test re-reads the reports and confirms each name and verdict). Counts at the time of writing: wolfram-algebra 45 checks, python-algebra 35, wolfram-field-theory 84, python-field-theory 70, wolfram-a4-report 52, python-a4-report 63, wolfram-pairing 101, python-pairing 66, wolfram-scope 15, python-scope 14, python-derive-eos 49, python-independent-numerics 28; all pass.

report	checks cited in this document
Revision/algebra/reports/wolfram-algebra.json	Clifford_relation, reality, symmetry_pattern, eta_in_author_order, coordinate_map, C_real_symmetric
Revision/algebra/reports/wolfram-algebra.json	C_squared_identity, C_gamma_real_antisymmetric, Gamma_diag, B_Hermitian, B_squared_identity, B_signature_8_8
Revision/algebra/reports/wolfram-algebra.json	S_Lorentz_algebra, S_preserves_C, S_commutes_with_Gamma, Pin44_irreducible_commutant_dim_1, Spin44_commutant_dim_2_chiral_projectors, chiral_halves_irreducible

report	checks cited in this document
Revision/algebra/reports/wolfram-algebra.json	chiral_halves_inequivalent_intertwiners_0, Spin_Pin_reflection_swaps_halves
Revision/algebra/reports/python-algebra.json	clifford_relation, clifford_relation_sympy, reality_signed_permutations, symmetry_pattern, C_equals_notebook_sigma16, C_real_symmetric_involution
Revision/algebra/reports/python-algebra.json	C_gamma_antisymmetric, chirality_diag, chirality_anticommutes, B_hermitian_involution_signature, S_lorentz_algebra, S_preserves_C_and_commutes_with_Gamma
Revision/algebra/reports/python-algebra.json	clifford_products_span_M16, pin_commutant_dimension_1, even_products_span_M8_plus_M8, spin_commutant_dimension_2, spin_halves_irreducible, spin_halves_inequivalent
Revision/algebra/reports/python-algebra.json	reflections_exchange_halves, fixture_comparison_gammas_json
Revision/theory/reports/wolfram-field-theory.json	metric_is_the_authors, vielbein_inverse, sqrt_det_g_is_cos_z, degenerate_at_H_0, christoffel_count, ricci_scalar
Revision/theory/reports/wolfram-field-theory.json	ricci_mixed_components, never_flat_for_H_positive, vielbein_postulate, omega_antisymmetric, omega_components, Omega_components
Revision/theory/reports/wolfram-field-theory.json	gamma_covariantly_constant, spin_curvature_equals_Riemann, gammaOmega_equals_3H_gamma_x8, gammaOmega_x4_terms_cancel, gammaOmega_divergence_form, L_real_C
Revision/theory/reports/wolfram-field-theory.json	L_total_divergence_to_unsymmetrised_C, L_spin_connection_drops_out_C, EL_Psibar_C, EL_Psi_C, adjoint_equation_is_Dirac_conjugate_C, Dirac_operator_explicit_C
Revision/theory/reports/wolfram-field-theory.json	block_form, evolution_form_C, nontriviality_2_dirac16complex00, Omega_vanishes_iff_a4prime_and_H_vanish, current_real_C, current_conservation_identity_C
Revision/theory/reports/wolfram-field-theory.json	charge_density_is_Krein_form_C, Lichnerowicz_identity_C, Majorana_Lg_total_derivative_grassmann, Majorana_Lg_commuting_control, T_vielbein_variation_closed_form_C, T_symmetric_part_Belinfante_C

report	checks cited in this document
Revision/theory/reports/wolfram-field-theory.json	Noether_identity_diffeomorphisms_C, Noether_identity_local_Lorentz_C, conservation_on_shell_general, T_diagonal_components_C, EMT_trace_C, kinetic_sum_on_shell_C
Revision/theory/reports/wolfram-field-theory.json	T_x4_x8_component_C, energy_exchange_equation, solution_matrix_square, exact_solution_x4_x8_C, exact_solution_nonlinear_homogeneous_C, good_sector_hermiticity_curved
Revision/theory/reports/python-field-theory.json	gammas_json_equals_python_construction, metric_from_vielbein_equals_SPEC, sqrt_det_g_equals_cos_z, christoffel_symmetric_metric_compatible, spin_connection_antisymmetric, vielbein_postulate
Revision/theory/reports/python-field-theory.json	Omega_x4_and_Omega_x8_vanish, covariant_constancy_D_mu_gamma_nu, gamma_mu_Omega_mu_equals_3H_gamma_x8, time_terms_cancel_hidden_term_survives, anticommutator_gamma_Omega_vanishes, divergence_of_sqrtg_gamma
Revision/theory/reports/python-field-theory.json	nontriviality_Omega_zero_iff_flat, curvature_nonzero_flat_only_formally, spinor_curvature_equals_riemann, lichnerowicz_contraction, lichnerowicz_identity_on_fields, commuting_lagrangian_real
Revision/theory/reports/python-field-theory.json	commuting_controls_not_vacuous, commuting_total_divergence_relation, commuting_euler_lagrange_psibar_variation, commuting_euler_lagrange_psi_variation, commuting_adjoint_equation_is_conjugate, commuting_current_conservation
Revision/theory/reports/python-field-theory.json	commuting_emt_symmetric, commuting_emt_conservation_on_shell, commuting_emt_conservation_negative_control, commuting_trace_on_shell, commuting_homogeneous_on_shell_rho_p, commuting_T_x4x8_homogeneous

report	checks cited in this document
Revision/theory/reports/python-field-theory.json	commuting_emt_equals_general_vielbein_variation_on_shell, commuting_emt_equals_vielbein_variation_diagonal, negative_control_majorana_grassmann_total_derivative, negative_control_majorana_commuting_contrast, energy_exchange_equation, exact_solution_family_x4_x8
Revision/theory/reports/python-field-theory.json	exact_nonlinear_homogeneous_solution
Revision/field_equations_a4/reports/wolfram-a4-report.json	metric_is_SPEC_section_1, sqrt_abs_det_g_is_cos_z, mixed_riemann_free_of_warp_and_a4, P1_direct_equals_minus_4_Einstein, P1_direct_equals_gkd_branch_monomials, P2_direct_equals_gkd_branch_monomials
Revision/field_equations_a4/reports/wolfram-a4-report.json	P3_direct_equals_gkd_branch_monomials, P1_trace_identity, P2_trace_identity, P3_trace_identity, P4_vanishes_pigeonhole, E1_divergence_free
Revision/field_equations_a4/reports/wolfram-a4-report.json	E2_divergence_free, E3_divergence_free, independent_components, evolution_factorises_a4pp_times_F, evolution_F_not_identically_zero, algebraic_identity_x1_plus_x5_minus_2x8
Revision/field_equations_a4/reports/wolfram-a4-report.json	x8_component_contains_no_a4pp, constraint_propagation_bianchi, conservation_components, einstein_components, einstein_null_energy_x8, einstein_no_vacuum_solution
Revision/field_equations_a4/reports/wolfram-a4-report.json	linear_member_equal_pressures, linear_member_vacuum_factor, C_properties_own_rep, gamma_mu_anticommutes_with_Omega_mu_no_sum, gravity_term_gamma_mu_Omega_mu, condensate_equation_x8_consistent
Revision/field_equations_a4/reports/wolfram-a4-report.json	condensate_S_constant, condensate_adjoint_equation, condensate_kinetic_tensor_diagonal, condensate_offdiagonal_are_three_gamma_bilinears, condensate_einstein_quadratic_U, condensate_diagonal_witness_exact
Revision/field_equations_a4/reports/python-a4-report.json	mixed_riemann_free_of_warp_and_a4, riemann_entries_laurent, P1_equals_minus_4_Einstein, P1_equals_gkd_branch_monomials, P2_equals_gkd_branch_monomials, P3_equals_gkd_branch_monomials

report	checks cited in this document
Revision/field_equations_a4/reports/python-a4-report.json	E1_divergence_free, E2_divergence_free, E3_divergence_free, evolution_factorises, evolution_F_not_identically_zero, algebraic_identity
Revision/field_equations_a4/reports/python-a4-report.json	constraint_and_x8_first_order, bianchi_x4, other_components_vanish, conservation_components, einstein_components, einstein_null_energy_x8
Revision/field_equations_a4/reports/python-a4-report.json	einstein_no_vacuum, linear_member_equal_pressures, linear_member_vacuum_factor, condensate_einstein_quadratic_U, json_lovelock_components, json_general_source_equations
Revision/field_equations_a4/reports/python-a4-report.json	json_einstein, json_linear_member, ownrep_clifford, ownrep_C_properties, ownrep_anticommutator_no_sum, ownrep_gravity_term
Revision/field_equations_a4/reports/python-a4-report.json	ownrep_condensate_S_constant, ownrep_condensate_adjoint, ownrep_condensate_kinetic_diagonal, ownrep_condensate_offdiagonal_three_gamma, ownrep_condensate_witness, authorT16_clifford
Revision/field_equations_a4/reports/python-a4-report.json	authorT16_C_properties, authorT16_anticommutator_no_sum, authorT16_gravity_term, authorT16_condensate_S_constant, authorT16_condensate_adjoint, authorT16_condensate_kinetic_diagonal
Revision/field_equations_a4/reports/python-a4-report.json	authorT16_condensate_offdiagonal_three_gamma, authorT16_condensate_witness, json_offdiagonal_coefficients, offdiagonal_coefficients_representation_independent
Revision/pairing/reports/wolfram-pairing.json	Gamma_properties, T1_kernel_field_equation, T1_linearity_argument, T1_Lagrangian_primordial_commuting, T1_Lagrangian_negative_controls_primordial_commuting, T1_field_equation_covariance_primordial_commuting
Revision/pairing/reports/wolfram-pairing.json	T1_Euler_Lagrange_map_primordial_commuting, T1_energy_momentum_primordial_commuting, T1_current_primordial_commuting, T1_general_potential_primordial_commuting, T2_mirror_is_isometry, T2_mirror_Lagrangian_commuting
Revision/pairing/reports/wolfram-pairing.json	T2_mirror_energy_momentum_and_current_commuting, T2_mirror_Euler_Lagrange_commuting

report	checks cited in this document
Revision/pairing/reports/python-pairing.json	T1.metric.commuting.lagrangian, T1.metric.commuting.negative_controls, T1.metric.commuting.euler_lagrange_map, T1.metric.commuting.emt, T1.metric.commuting.pair_total_emt_zero, T1.metric.commuting.current
Revision/pairing/reports/python-pairing.json	T1.metric.commuting.general_potential, T2.metric.commuting.euler_lagrange_map, T2.metric.commuting.emt, T2.metric.commuting.current, T2.metric.commuting.S_odd, Q.one_particle_Krein_inertia_proof
Revision/pairing/reports/wolfram-pairing.json	Q_one_particle_Krein_inertia_real_frequencies
Revision/theory/reports/wolfram-scope.json	boosted_frame_reproduces_metric, boosted_frame_canonical_connection, boosted_frame_gammaOmega_formula, boosted_frame_gammaOmega_vanishes, boosted_frame_curvature_nonzero, rescaling_removes_the_connection_term
Revision/theory/reports/wolfram-scope.json	rescaled_equation_quadratic_potential, gammaOmega_blind_to_the_deflation, connection_free_lagrangian_same_equations, spin_connection_in_the_energy_momentum_tensor, good_sector_hermiticity_up_to_the_brane_flux
Revision/theory/reports/wolfram-scope.json	good_sector_x8_independent_modes_without_boundary_condition, extra_time_growth_rates_unbounded, commuting_field_energy_unbounded_below
Revision/theory/reports/python-scope.json	boosted_frame_reproduces_metric, boosted_frame_canonical_connection, boosted_frame_gammaOmega_formula, boosted_frame_gammaOmega_vanishes, boosted_frame_curvature_nonzero, rescaling_removes_the_connection_term
Revision/theory/reports/python-scope.json	rescaled_equation_quadratic_potential, gammaOmega_blind_to_the_deflation, connection_free_lagrangian_same_equations, spin_connection_in_the_energy_momentum_tensor, good_sector_hermiticity_up_to_the_brane_flux

report	checks cited in this document
Revision/theory/reports/python-scope.json	good_sector_x8_independent_modes_without_boundary_condition, extra_time_growth_rates_unbounded, commuting_field_energy_unbounded_below
Revision/dark_sector/dirac16complex00/reports/python-derive-eos.json	observer_N2_weff_identity, M4_tangent_equals_unite, M4_never_phantom, M5_crosses_minus_1, M5_without_ghost_no_crossing, condensate_ratio_equals_unite_constant_w
Revision/dark_sector/dirac16complex00/reports/python-derive-eos.json	linear_member_einstein_phantom_ratio, expansion_inferred_w_einstein

15 What is not claimed

- **No quantisation.** dirac16complex00 is not quantised; its energy-momentum tensor is a classical bilinear. The operator statements (Krein space, Fock space, normal ordering) belong to dirac16complex.
- **The energy is not bounded below, the charge is indefinite.** The record decides the energy question negatively: already for $U = 0$ in the good sector there are positive-frequency solutions of negative energy, and the energy density $-5|c|^2$ of the example of section 9 is unbounded below as $|c|$ grows; the charge $\int \cos z \, \Phi^\dagger B \Phi$ is indefinite and only locally conserved (it is constant only if no charge flows through the brane $z = \pi/2$; see the item on the brane below). The descriptions "semi-classical" and "analogue of Dirac's 1928 wave function" are interpretations: that wave function has a positive conserved density, this field does not.
- **Frame dependence of the gravitational term.** $\gamma^\mu \Omega_\mu = 3H\gamma^{(x_8)}$ is the value for the diagonal vielbein and the variables Φ; it vanishes in the boosted frame of section 6 and is removed by the rescaling $\Phi = \sin^{-1/2} z \, \chi$. Only Ω_μ itself (curvature) and the frame factors are frame-independent; non-triviality [2] is claimed in this exact sense.
- **No 3-space observer equation of state, no CPL fit, no comparison with supernova data.** Section 10 gives the ratios p/ρ of the 8-dimensional tensor only; Hypothesis00 (a time-varying dark-energy or dark-matter equation of state from dirac16complex00) is investigated in the separate dark-sector record `Revision/docs/DARK_SECTOR_HYPOTHESES`, not here (its verdict is summarised in section 10: it establishes neither hypothesis). For the homogeneous condensate the ratio w is constant in x_4.
- **No general solution, no well-posed Cauchy problem.** The exact solutions of section 5 are test points; the field equation is not solved in general, and no existence statement for the nonlinear Cauchy problem is made beyond the first-order evolution form. For data that depend on the extra times the problem is ill-posed in Hadamard's sense (section 7), and the good sector needs a boundary condition at $z = \pi/2$, which this record does not impose.
- **No realisation of the deflating member by a specific condensate.** The theorem of section 12 shows that a homogeneous condensate allows only $a_4 = AHx_4 + a_0$ and gives the

conditions the source must satisfy; the sign of A (deflation for $A > 0$) is not selected by the equations; it is not shown that a specific dirac16complex00 condensate meets those conditions with definite κ, Λ, m, λ and $A > 0$, and no non-homogeneous source with $k_1 \neq k_5$ (which would drive a_4'') is constructed.

- **The x_8 dependence.** The metric of section 1 is an exact solution only with an x_8-independent, isotropic source with vanishing off-diagonal components; a dirac16complex00 configuration that depends on x_8 is not a consistent source for this metric.
- **The brane.** The patch end $z = \pi/2$ is a degenerate surface of the metric ($g_{88} = 0$, $\sqrt{|g|} = 0$) at finite proper distance; no junction condition is derived, the mirror construction used by T2 is an assumption, and the boundary term $\sin z\, u^\dagger M_8 v$ of the hidden direction does not vanish there.
- **Pairing.** The pairing theorems of section 13 derive no creation process, rate or amplitude for universes of masses $\{+m, -m\}$; they are correspondences between solutions, in a fixed gravitational field.
- **Notebook conventions.** The canonical spin connection used here is the one with $\omega_{\mu ab} = \eta_{ac}\omega_\mu{}^c{}_b$; the notebook's contraction of the mixed components (which lacks a metric factor) is not used, and its Majorana-type L_g serves only as the negative control of section 4.

16 Reproduction

From the repository root, in this order (each verifier writes its report and exits with code 0 only if every check passes; two runs give byte-identical outputs):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/theory/wolfram/verify_field_theory.wls
python Revision/theory/python/check_field_theory.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/pairing/wolfram/verify_pairing.wls
python Revision/pairing/python/check_pairing.py
wolframscript -file Revision/theory/wolfram/verify_scope.wls
python Revision/theory/python/check_scope.py
```

Re-run for this document (2026-10-01). The eight verifiers of the first edition were run in this order on a scratch copy of the Revision inputs. Run times on the development machine: algebra 4 s (Wolfram) and 2 s (sympy); theory 1029 s (Wolfram) and 115 s (sympy); a_4 equations 15 s and 4 s; pairing 82 s and 147 s. After the review of 2026-10-01 the sympy theory checker, the pairing verifiers, the a_4 verifiers and the new scope verifiers were run again on the working tree: every check passes, and the outputs of two runs are byte-identical. The comparison section `comparison_with_wolfram` of `Revision/theory/reports/python-field-theory.json` was regenerated against the current Wolfram record (84 checks): 32 of 32 formula records and 63 of 63 check pairs agree; its parser now reads the prose form of `energy_exchange` in the current `Revision/theory/field-theory.json`, and the five prose records that it did not compare before are compared by re-deriving

their formulas. The sympy theory checker now exits with an error when the comparison disagrees (103 s for the regenerated report; the scope verifiers 17 s and 1 s). The reports cited here are the files of the working tree.

The two dark-sector programs whose reports section 10 cites (their place in the full order and their run times are given in section 12 of `Revision/docs/DARK_SECTOR_HYPOTHESES`):

```
python Revision/dark_sector/dirac16complex00/python/derive_eos.py
python Revision/dark_sector/dirac16complex00/python/independent_numerics.py
```

The document is built and checked with (the first command on one line)

```
python scripts/build_provenance_pdf.py Revision/docs/DIRAC16COMPLEX00_FIELD_THEORY.md
       --developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_dirac16complex00_field_theory_publication.py -v
```

The builder makes the .tex twice and the PDF in two fresh directories, requires byte identity and a warning-free LaTeX log, and verifies the PDF against its entry in Revision's own registry `Revision/pdf-specifications.json`. The publication test checks that the .tex is the builder's output for this .md, that the PDF is registered with its page count and sha256, that the formula blocks of sections 5, 11 and 12 are the ones generated from `Revision/theory/field-theory.jso n` and `Revision/field_equations_a4/a4-equations.json`, and that every check cited in section 14 exists in its report with a passing verdict.